Unsupervised Wrap-Discontinuity Repair in Wavetable Synthesis via Hybrid GA–PPO Meta-Search

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Code (ReelSynth engine + DenoiseOpt meta-search): https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

Mismatched wavetable endpoints leave a wrap-join click at the seam on every period. We frame unsupervised repair of that local discontinuity around a procedural cliff-withheld ideal sibling for self-supervised seam scoring plus mid-cycle identity to the cracked engine: search never sees paired studio-clean targets. Classical raised-cosine fades, virtual-analog seam operators, and corrupt-to-corrupt Noise2Noise are strong baselines on some boards. Because no single operator fits every engine family, **DenoiseOpt** applies a hybrid genetic-algorithm (GA), proximal policy optimization (PPO), and population-based training (PBT) meta-loop to search compact repair operators under the blended residual R_{blend} and latency-aware objective J. This is an application of known search tools to a crisp DSP problem. Across five holdout seeds, Ours sits below frozen Noise2Noise (0.9708 ± 0.0014 vs 0.9767 ± 0.0015) and well above Dual Cosine (0.542 ± 0.038). On twenty multi-family boards the mean-of-means edges Noise2Noise (0.9337 ± 0.0028 vs 0.9308 ± 0.0045; mean win fraction 0.47). Matched-5k outer-loop re-search under $R_{\text{blend}} + J$ across three seeds keeps hybrid ahead of Random/CMA-ES/TPE/Aging/REINFORCE (0.9748 ± 0.0018). On hard wrap cliffs classical fades collapse while searched operators hold. Transfer benches reuse the same wrap-discontinuity protocol on other periodic signals as stress tests, not domain diagnosis claims.

Keywords wavetable synthesis; wrap discontinuity; unsupervised repair; DenoiseOpt; seam restoration; Noise2Noise; neural architecture search; reinforcement learning; genetic algorithms

1 Introduction

Wavetable oscillators store one period and wrap the read pointer at the end of the table. When those endpoints disagree, each wrap injects a discontinuity that listeners hear as a period-locked click.

Wavetable synthesis in brief. A wavetable synthesizer stores a single cycle of a waveform (the wavetable) and produces continuous tone by looping a read pointer through that table. Pitch is set by how fast the pointer advances. Timbre comes from the stored shape (and, in multi-table banks, from morphing across tables). Tiling the stored period yields a sustained note under playback. If the endpoints disagree ($x[0] \neq x[L-1]$), every wrap is a discontinuity at the wrap join (the seam), heard as a click locked to the fundamental. Figure 1 shows one stored period, tiled playback with wrap marks, and a zoomed mismatched wrap. The same geometry appears whenever a short loop is tiled for listening or scoring.

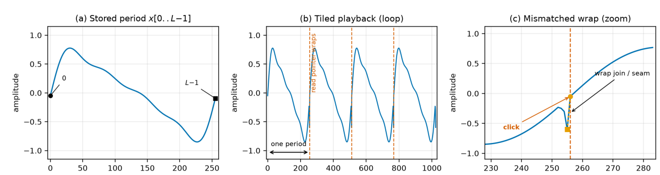

Figure 1: Wavetable synthesis primer. (a) One stored period $x[0..L-1]$ (L=256). (b) The same period tiled for continuous playback. Dashed lines mark read-pointer wraps. (c) Zoom at a mismatched wrap join (seam), where the endpoint cliff produces an audible click.

Terminology. Key names used below. A **seam** (wrap join) is the local end/start junction where endpoint mismatch produces the period-locked click: the seam is local, while the signal is the whole waveform. **Repair operator** Θ means a searchable local edit at that seam (legacy: bake cell), and a tiny network inside Θ is a **residual operator**. **FitCell** continuously fits one candidate Θ under residual loss (Appendix A). Scores R and R_{blend} are unsupervised: prolonged R is debug-only, and R_{blend} drives primary selection together with the latency-aware objective $J = R_{\text{blend}} - \lambda \cdot \text{latency\_norm}$ (Section 3.3). **MoE** soft-gates bake experts, and **PBT** mutates elites in the outer loop. The **outer loop** (meta-search) runs

GA plus PPO over repair-operator graphs rather than end-to-end training of one large network. A **holdout** freezes the evaluation set or seed for reported scores.

Unsupervised, as used here. Search never trains on paired studio-clean audio. The ideal sibling $r^\star$ is a procedural cliff-withheld twin of the cracked engine (self-supervised geometry for scoring), not a human-labeled clean recording. Operators are ranked under R_{blend} without requiring labeled clean targets at search time.

We treat the task as *cycle-local seam restoration*: edit a neighborhood of the wrap join (the seam), and optionally a small residual operator over the period, so multi-period playback matches an ideal reference that never opens the wrap cliff. Agreement uses a discontinuity-weighted blend R_{blend} (seam fidelity vs the ideal, plus mid-cycle identity vs the cracked engine) under a latency-aware objective J (Section 3.3).

Four roles. Keep these distinct (Figure 2 and Section 3.4):

1. **Ideal sibling** $r^\star$: cliff withheld. Scoring target only. Not a method.
2. **No-bake**: unrepaired cracked engine x, scored vs $r^\star$. A do-nothing reference.
3. **Dual Cosine**: classical raised-cosine end fade. Classical baseline.
4. **Ours (DenoiseOpt)**: searched repair operator from the hybrid outer loop.

Corrupt-to-corrupt **Noise2Noise** is a fifth comparator: the primary neural gate under the locked R_{blend} protocol (Section 5.2). It is not shown in Figure 2, which illustrates roles on one hard sine-wrap tile.

Classical virtual-analog tools already address wrap energy locally (BLIT/BLEP [1], PolyBLEP [3], BLAMP [2]). We score cycle-local bake versions of those operators beside Dual Cosine (Section 5.14). Label-free restoration such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4, 5]. On the search side, neural architecture search (NAS), aging evolution, PBT, covariance matrix adaptation evolution strategy (CMA-ES), and evolution-guided policy gradients motivate a hybrid outer loop: GA for diversity, PPO for discrete architecture edits, and PBT-style exploit–mutate on elites [16, 19, 21–23].

Key findings. Under the locked v10.1 R_{blend} protocol with multi-seed evaluation (primary story: Table 6):

- On the primary holdout, the hybrid champion sits just below the frozen Noise2Noise gate on every seed (0.971 ± 0.001 vs 0.977 ± 0.001) and well above Dual Cosine (0.542 ± 0.038).
- On twenty multi-family boards, Ours mean-of-means edges Noise2Noise (0.934 ± 0.003 vs 0.931 ± 0.005; mean win fraction 0.47 across seeds).
- On hard wrap cliffs, classical fades collapse while searched operators and Noise2Noise hold (Section 5.10).
- On Factory+FX and AKWF corpora under R_{blend}, Noise2Noise leads Ours; both still beat Dual Cosine.

Compact searchable repair operators remain competitive when classical fades fail and need no paired clean targets at search time. Whole-curve prolonged R near 0.991 is a superseded debug score, not a current holdout result under R_{blend}.

Transfer datasets. Transfer boards apply the *same* wrap-discontinuity protocol to other short periodic signals (Section 4.2): open a cliff at the period join, score against a no-cliff ideal sibling. They are mechanistic stress tests of wrap geometry under that protocol, not evidence of general domain diagnosis. Boards: **CWRU**, **MFPT**, and **Paderborn** K001 (one shaft revolution as a period), **MIT-BIH** and a **PTB-XL** 500 Hz subset (one heartbeat as a period), plus **synthetic CNC** and **synthetic PMU/AC** pilots when real downloads are walled (Table 15).

Scope. Cycle-local wavetable seam repair for DSP / synthesis venues (DAFx / AES / arXiv cs.SD). Formal wrap-closure proofs, search-convergence theorems, and speech/music enhancement SOTA are out of scope; caveats live in Section 9.

1.1 Contributions

1. **Problem framing.** Wrap discontinuity as cycle-local seam restoration under tiled evaluation, with explicit R_{seam}, R_{body}, and primary R_{blend}.
2. **Ideal-sibling protocol.** Unsupervised scoring via a procedural cliff-withheld twin plus mid-cycle engine identity, without paired studio-clean targets.
3. **Applied hybrid search.** DenoiseOpt applies GA–PPO(+PBT) meta-search to compact repair-operator graphs for this DSP task, scored by R_{blend} and ranked by J.
4. **Open multi-seed evaluation.** Frozen Noise2Noise gate, multi-family and wavetable-native boards, hard-cliff strata, and named wrap-protocol stress-test benches with released code and artifacts.

2 Related Work

We group prior work by mechanism. **Used** citations ground methods we build on or compare against under our protocol. **Screened** citations mark prior work we reviewed and ruled out (wrong domain, cost, or objective). We cite those papers for positioning only. They are not building blocks of DenoiseOpt.

Wavetable / virtual-analog foundations and discontinuity control. Foundational wavetable and virtual-analog (VA) oscillator work frames alias-free classic waveforms and wrap/corner discontinuities via BLIT/BLEP [1], PolyBLEP [3], and BLAMP [2]. We treat those OA papers as the wavetable/VA lineage for this seam task and score cycle-local

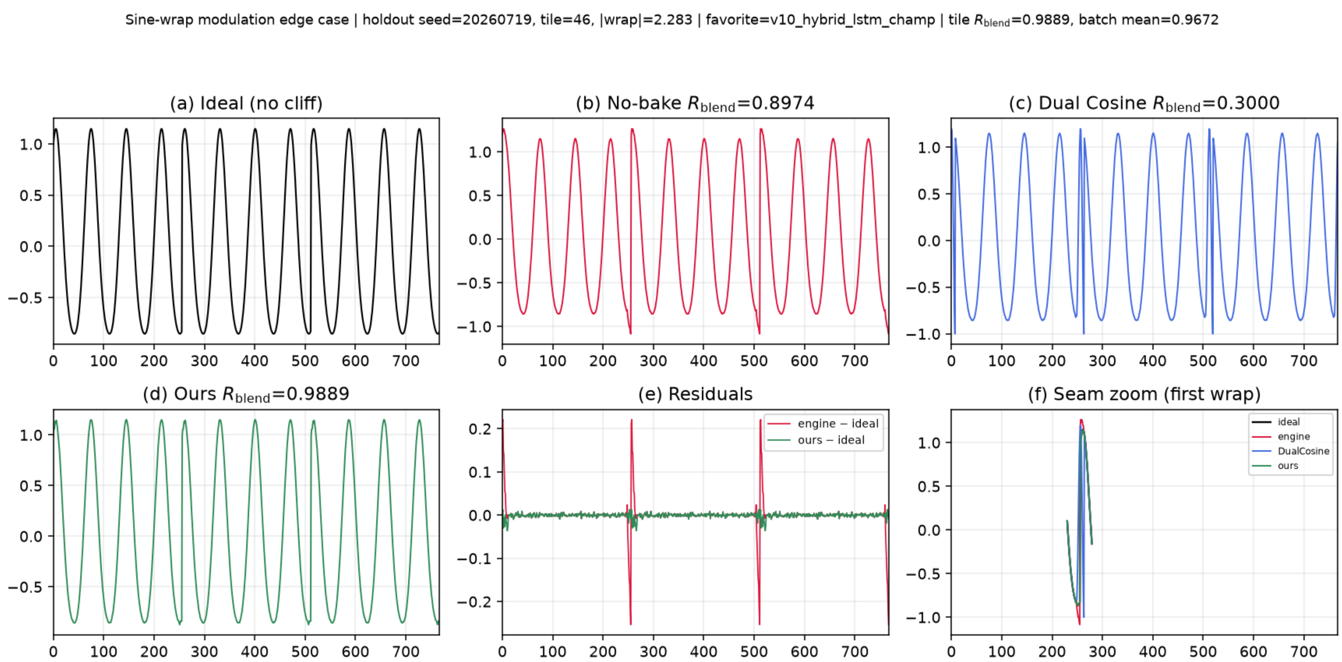

Figure 2: Hard sine-wrap holdout tile (seed `20260719`). (a) Ideal sibling $r^\star$. (b) No-bake cracked engine. (c) Dual Cosine fade. (d) Ours (searched bake). (e)–(f) Residual and seam zoom. Accompanying JSON scores use locked R_{blend} (α=0.7; tile 46: no-bake 0.897, Dual Cosine 0.300, Ours 0.989; Table 6).

approximations beside Dual Cosine (Section 5.14). DDSP frames DSP modules as composable blocks [6]. We reuse that framing for small bake/FIR/MLP seam operators at cycle scale, rather than end-to-end vocoders.

Label-free restoration. Noise2Noise learns restorers from corrupted observations alone [4,5]. Mixture-invariant training ranks unsupervised systems without stem labels [29]. We train a corrupt-to-corrupt seam CNN on the same generator as the primary neural baseline (Section 5). Full Demucs / DiffWave stacks are screened: domain and bake-budget mismatch [34, 37].

Compact waveform operators. Wave-U-Net, Conv-TasNet, dual-path, and attention audio models supply block priors we shrink to bake width [8, 11–15, 30]. They supply block priors rather than serving as frozen full-scale separators inside every meta trial.

Search and hybrids. Neural architecture search (NAS) taxonomies, reinforcement-learning (RL) controllers, progressive and latency-aware NAS, aging evolution, population-based training (PBT), CMA-ES, and evolution-guided policy gradients motivate applying a hybrid outer loop to this DSP task [16–19, 21–24, 27, 28]. Mixture-of-experts (MoE) soft gates motivate routing among small heterogeneous operators under one residual score [25]. Matched outer-loop bake-offs (Random, CMA-ES, REINFORCE, aging evolution, TPE vs hybrid) sit in Appendix B. We do not claim a new foundational NAS/RL algorithm.

Used vs screened (short). Used anchors include VA discontinuity control [1–3], unsupervised restoration [4, 5, 29], DDSP [6], compact waveform operators as above, and NAS/RL/PPO/PBT/CMA-ES/ERL/MoE as above. Screened as contrast only: MAML [32], DARTS [33], Demucs-scale enhancers [34, 36], DiffWave [37], and SEGAN-style full GAN loops [38]. Claims stay inside cycle-local seam restoration under a declared compute budget.

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the cracked single-period wavetable (engine input with open-wrap cliff). Let $\mathscr{H}$ be the space of repair-operator configurations (discrete operator graph plus continuous bake parameters). The repair operator (bake operator Θ) is

$$\Theta : \mathbb{R}^L \times \mathscr{H} \to \mathbb{R}^L, \qquad y = \Theta(x;h), \quad h \in \mathscr{H}. \tag{1}$$

When continuous parameters alone are emphasized we write θ for the continuous slice of h and $y = \Theta(x;\theta)$. $r^\star \in \mathbb{R}^L$ is the procedural ideal sibling defined below. Tiling length N (typically N=16) forms $y_{\text{tiled}} = \text{Tile}(y,N)$ and $r^\star_{\text{tiled}} = \text{Tile}(r^\star,N)$. Seam width W (code: `SEAM_W`, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0,1]$ is the prolonged residual score (1 = best). These symbols follow a different convention from Noise2Noise clean/noisy pairs.

Table 1: Core symbols used in Methods.

Symbol	Meaning
L	Cycle length (samples per period)
x	Cracked engine wavetable period (cliff on)
$y = \Theta(x;h)$	Baked cycle after seam ops / residual operator
Θ	Repair operator (bake operator) $\mathbb{R}^L \times \mathscr{H} \to \mathbb{R}^L$
$h \in \mathscr{H}$	Repair-operator configuration (ops + continuous θ)
$r^\star$	Ideal sibling (same seed, cliff withheld)
G	Procedural period generator (cliff on/off)
N	Prolong tiling count
W	Seam neighborhood width
R	Prolonged residual score in $[0,1]$ (1 = best)

3.2 Problem setup: cracked period and ideal sibling

Problem. Given a cracked period x, choose a bake configuration h so that the repaired cycle $y = \Theta(x;h)$ matches a no-cliff ideal under prolonged tiling.

Ideal sibling. Let $G(\text{seed}, \text{cliff})$ denote the procedural generator used by the CUDA FitCell batch maker (FitCell = inner continuous-parameter fit of one candidate Θ; Algorithm 5): one draw of frequency, phase, and mid-cycle content, with an optional open-wrap cliff of width W at both ends. For any seed,

$$r^\star = G(\text{seed, cliff=off}), \qquad x = G(\text{seed, cliff=on}). \tag{2}$$

Engine and ideal therefore share the mid-cycle content of one draw and differ by the seam cliff (plus any seam-boosted noise applied only on the engine path). The ideal is *not* a latent studio ground truth and is *not* a method output. For an arbitrary cracked period from this family, the sibling is obtained by replaying the same seed with the cliff withheld. That is exactly the pair returned by `make_batch`. This sibling relationship makes unsupervised ranking possible without paired studio recordings [4,5]. In this paper, *unsupervised* means: no human-labeled clean audio and no studio-clean targets in the search loop. The ideal sibling is a procedural self-supervised scoring twin (cliff withheld), not a perceptual ground truth.

3.3 Blended residual score and latency-aware objective

Primary metric. After tiling each cycle N times, let y be the repaired output, $r^\star$ the ideal sibling, and x the cracked engine. Define the wrap-neighborhood mask $\mathscr{S}$ of width W=8 at each period head/tail and the complementary mid-cycle body mask $\mathscr{B}$. Component scores use the unit clamp of relative RMS:

$$R(a,b;\mathscr{M}) = \text{clamp}\left(1 - \frac{\text{RMS}\big((a-b)\odot \mathbf{1}_{\mathscr{M}}\big)}{\text{RMS}\big(a\odot \mathbf{1}_{\mathscr{M}}\big)+\varepsilon}, 0, 1\right). \tag{3}$$

Then $R_{\text{seam}} = R(r^\star, y; \mathscr{S})$ (seam fidelity vs the ideal) and $R_{\text{body}} = R(x, y; \mathscr{B})$ (mid-cycle identity vs the engine). The primary reported score is

$$R_{\text{blend}} = \alpha R_{\text{seam}} + (1-\alpha) R_{\text{body}}, \quad \alpha = 0.7. \tag{4}$$

Whole-curve prolonged R remains a debug / superseded key only.

Roles of the three quantities.

Quantity	Role	Used for
R_{blend}	Primary quality metric	Tables, figures, reporting
J	Latency-aware search objective	Champion selection
N2N gate	Frozen neural benchmark	Holdout pass/fail

Search objective. Candidates maximize latency-aware

$$J = R_{\text{blend}} - \lambda \cdot \text{latency\_norm}, \quad \text{latency\_norm} = \frac{\log(1+t_{\text{ms}})}{\log(1+50)}, \quad \lambda = 0.02. \tag{5}$$

FitCell minimizes $1 - R_{\text{blend}}$. Selection and champion updates use J. A champion must also strictly beat the frozen corrupt→corrupt Noise2Noise holdout R_{blend} (primary baseline).

Every scored row applies some Θ to x and is evaluated under (4). The ideal sibling is the seam scoring target. Body identity uses the engine.

Remark 1 (Noise2Noise primary baseline). `SeamN2N` (≈53.5k) is the primary neural comparator and champ gate. Search may discover `n2n_unet` operators at the same scale. TinyUNet1D `unet` is a different, smaller block [4,5].

3.4 Seam bake operator and classical baselines

A **repair operator** $\Theta(\cdot;h)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual operators (MLP/FIR/U-Net-lite/attention-lite/LSTM-lite) and optional mixture-of-experts (MoE) soft gates over experts (Appendix A, Algorithms 2–4). Bake parameters sit in a bounded box; architecture strings are discrete graphs over those ops. Separate classical controls apply cycle-local BLIT/BLEP, PolyBLEP, and BLAMP residual corrections at the wrap on the cracked engine tile: they share the DualCosine / FIR interface and are not a full oscillator rewrite. Roles in plain language: BLIT/BLEP and PolyBLEP correct a *value* step at wrap, BLAMP corrects a *slope*/corner jump, and DualCosine fades endpoints without a BLEP residual. Section 5.14 scores them.

Four roles (keep separate). Four objects must stay distinct (Figure 2). The **ideal sibling** $r^\star$ withholds the cliff and is a scoring target only (panel (a)). **No-bake** is the unrepaired passthrough $\Theta_{\text{no-bake}}(x) = x$: the cracked engine scored against $r^\star$ (panel (b)). It leaves the engine unchanged and must not be confused with $r^\star$, residual-net skips, or learned identity maps. **DualCosine** is the classical raised-cosine end fade on x (panel (c)). It appears both as a classical board row and, separately, as a PPO advantage baseline (Remark 2). **Ours** is the searched bake $y = \Theta(x;h^\star)$ from the hybrid outer loop (panel (d)).

On mild cliffs, residual mass sits far below ideal RMS under the unit clamp of (4), so no-bake R_{blend} can look high even

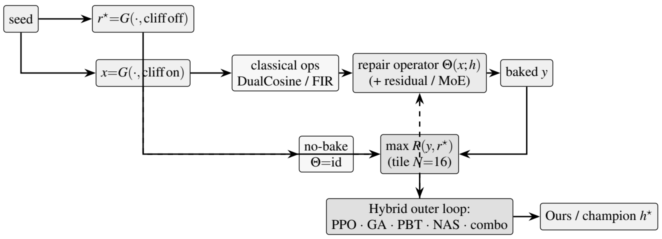

Figure 3: DenoiseOpt overview (before component deep-dives). Same-seed generator G yields ideal sibling $r^\star$ (cliff off) and cracked engine x (cliff on). Repair operator $\Theta(x; h)$ may include DualCosine/FIR and optional residual/MoE experts. No-bake is the identity control $x \mapsto x$. Outer objective is maximize absolute prolonged $R(y, r^\star)$. DualCosine centering (not shown) is PPO advantage shaping only. Dashed feedback: architecture proposals update the operator. Pseudocode in Appendix A.

when the wrap click remains audible. Treat that number as a do-nothing passthrough reference rather than a perceptual wrap-quality score. Released JSON may still use the legacy key `identity` for no-bake. Manuscript prose uses *no-bake*.

Classical (non-AI) comparison board. Learned / searched methods are always ranked against the full classical board on absolute R (same $r^\star$): no-bake, DualCosine / cosine fade, `seam_fir3`, classic quadratic, hann/crossfade/soft fades, detrend/ensemble, poly seam, and cycle-local BLIT/BLEP / PolyBLEP / BLAMP where scored. When tables show ΔR vs DualCosine, that is a reporting gap vs one classical bake, not the definition of reward.

Remark 2 (DualCosine centering is PPO advantage only). The outer objective remains (5): maximize absolute $R(y, r^\star)$. In the PPO branch only, we optionally form a centered advantage

$$\rho = R - R_{\text{DualCosine}}, \tag{6}$$

so early-search advantages are roughly zero-mean. This does *not* change selection, FitCell loss, champion ranking, or the matched 5k meta-approach objective. All of those still maximize absolute R. DualCosine centering is soft relative shaping for the actor–critic, not a hard absolute ranking target.

3.5 Search pipeline

Figure 3 summarizes the residual-scored hybrid loop. Each outer iteration proposes a bake configuration h, runs FitCell (Appendix A, Algorithm 5) to fit continuous parameters under loss $1 - R$, evaluates prolonged R vs $r^\star$, and updates the population / PPO policy. Pseudocode for GA, PPO, PBT, and the full hybrid rotation is in Algorithms 6–9.

3.6 Search space and inner fit

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth, residual operator type (none, MLP, FIR, U-Net-lite, attn-lite, LSTM-lite), optional MoE expert count $K \in \{2, 3, 4\}$, and continuous θ in the CUDA runner box bounds. The searchable bake vocabulary includes a width-capped 1–2 layer LSTM block over the seam window or full cycle (distinct from the fixed supervised seq-LSTM SOTA baseline). The live hybrid tag is `PPO+GA+PBT+NAS+depth+MoE` with LSTM enabled in the block graph.

FitCell draws sibling batches via G, applies $\Theta(\cdot; h)$, and steps Adam on $1 - R$ with early stopping on plateau (Algorithm 5).

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection and FitCell are always by absolute residual R (maximize toward the ideal sibling, loss $1 - R$).

Near-ceiling scores need reward shaping. On the sine+cliff generator, no-bake already sits near $R \approx 0.97$, and strong classical / searched bakes live in a narrow band toward 1 (often 0.99–0.991). Absolute gaps of order 10^{-3}–10^{-2} are real but tiny as raw PPO/GA signals. Without shaping, advantages collapse. Hyperparameters (clip, entropy, mutation rate, population size, fit steps, learning rate) then decide whether the outer loop can climb the last percent. We therefore treat reward shaping and search hyperparameters as first-class tunables alongside bake architecture. In the present hybrid, PPO uses a searchable shaping mode among {`abs_r`, `vs_dualcosine`, `vs_nobake`, `neglog_gap`} (PBT-mutated, default $\rho = R - R_{\text{DualCosine}}$ per Remark 2). PBT mutates fit / PPO hyperparameters and reward mode in-population. Plateau adaptation retunes branch mix and depth/MoE bias when champions stall. The matched 5k meta-approach comparison (Section 3.8) is the sole GPU experimentation vehicle for outer-loop family, HP, and reward-shaping sweeps under a shared FitCell budget.

3.8 Meta-learning approach comparison

Beyond the hybrid outer loop, we compare five additional outer-loop families under a matched 5,000-evaluation budget, three search seeds (`1902771841`, `2026072701`, `2026072702`), and the same FitCell / $R_{\text{blend}}+J$ protocol (LSTM and xLSTM included in the searchable bake vocabulary). Table and figure legends use the short display names below. Artifact folders keep stable code keys (see the nomenclature):

- **Random NAS** (`random`): independent uniform draws over bake graphs and fit hyperparameters (control).
- **Cont. CMA-ES** (`cmaes`): diagonal covariance-matrix adaptation over a continuous encoding of depth, width, operator kind, block logits, and fit hyperparameters [24].
- **Arch REINFORCE** (`reinforce`): vanilla policy gradient over discrete architecture-edit actions (no clipped surrogate). Contrast to PPO [17, 19].
- **Aging evolution** (`aging_evo`): regularized evolution with oldest-member replacement after tournament parent

Table 2: Hybrid search hyperparameters (RTX 3090 DenoiseOpt CUDA search-runner defaults). Co-tuned with reward shaping under near-ceiling R (Section 3.7).

Parameter	Value
Population size n	12
GA tournament k	3
GA crossover fraction p_c	0.5 (else clone elite parent)
GA mutation	≥ 1 mutate after inherit (+50% second mutate)
PPO clip ε	0.2 (default, searchable $\{0.1, 0.2, 0.25, 0.3\}$)
Adam learning rate (fit)	3×10^{-3} (default, PBT-searchable)
Adam learning rate (policy)	3×10^{-4}
Entropy coefficient (PPO)	0.02 (default, searchable ≈ 0.005–0.06)
PBT exploit / mutate	elite fraction 0.25, multi-step arch+hp mutate
MoE routing	`sequential` or `moe_parallel` when enabled
Plateau boredom threshold	1000 iters without champ improve
Cycle length L / tiles N / seam W	256 / 16 / 8
Fit steps / batch (default)	24 / 48
Algo tag	`PPO+GA+PBT+NAS+depth+MoE`

selection [20, 21], distinct from the hybrid's non-aging tournament GA branch.

- **TPE Bayes NAS** (`tpe`): lightweight tree-structured Parzen estimator over discrete bake categoricals in the spirit of practical Bayesian optimization [26].

We also run **Ours (hybrid GA–PPO)** (`hybrid_lstm`): the multi-branch hybrid loop for 5,000 evaluations with LSTM/xLSTM in the block vocabulary and online reward-mode / fit–PPO hyperparameter co-tuning, as the proposed matched-budget arm. Results appear in Appendix B (Table 17, Figures 21–22). Healed-sample overlays appear in Figure 7.

Remark 3 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual operators) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

3.9 Hyperparameters

Table 2 lists the CUDA search defaults used for the 5k-gate freeze reported in Results. Under the near-ceiling residual regime (Section 3.7), these outer-loop and fit hyperparameters are co-tuned with architecture. PBT and plateau adaptation continue to mutate several of them online. Companion listings: Algorithms 1–9 (Appendix A) and the pseudocode notes.

4 Experiments

Locked evaluation protocol. Evaluation follows the v10.1 evaluation protocol. Primary metric is R_{blend} (α=0.7); search objective is J as in (5) (λ=0.02). The champ gate is to strictly beat frozen Noise2Noise corrupt→corrupt holdout R_{blend}. Secondary keys cover SNR, SDR, wrap-jump, and seam-local edge RMSE, with debug keys for pure R_{seam} and whole-curve R. Holdout evaluation uses five seeds (`20260719`–`20260723`); search uses three seeds (`1902771841`, `2026072701`, `2026072702`); N2N train seed is `424242`. We report the hybrid_lstm freeze plus holdout / multi-family / Factory+FX / AKWF / transfer benches. PESQ/STOI stay out of scope on non-speech cycles, and LibriSpeech / MUSDB are not used. Venue: DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD).

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used by the CUDA search runner (`make_batch`). Cycles are length L=256. A two-harmonic sine is drawn with frequency $\sim \mathscr{U}(1,4)$ and phase $\sim \mathscr{U}(0,2\pi)$, then an open-wrap seam cliff of amplitude $\pm\mathscr{U}(0.08, 0.43)$ is applied over `SEAM_W`=8 samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, 0.15× in the mid-cycle) forms the engine input. The ideal withholds the cliff and serves only as scoring target $r^\star$, never as a method output, while no-bake leaves the cracked engine unchanged and scores that unrepaired x against the same $r^\star$. Prolonged residual R tiles each cycle N=16 times before the RMS ratio. Holdout seed `20260719` is distinct from hybrid search seed `1902771841`. The search campaign draws fresh i.i.d. batches from this generator and never trains on the frozen holdout tensors: there is no supervised clean/noisy training split of studio audio.

Multi-family diversity ($\geq$ 20 waveforms). Beyond the single sine+cliff holdout, Phase 3a scores methods on **20** generative draws: ten family variants (`sine_cliff`, `harmonic_fft`, `am_fm`, `nonlinear`, `combo`, `triple_mix`, `extreme_overlay`, `open_wrap_bias`, `soft_noise`, `wide_seam`) × two seeds (`20260719` and `20260720`), batch 64 each (128 cycles per family, 1,280 total). These Python generative variants stress harmonic / AM–FM / nonlinear / overlay / open-wrap behavior of `make_batch` and remain the primary SOTA matrix. Separately, `export_sound_bench_tiles` exports **20** Rust `sound_bench` engine/ideal pairs (10 families × 2 seeds, L=256, byte-aligned with `generate_sound` / `generate_sound_ideal`) scored under the same residual geometry as a secondary transfer block (Table 14). Every method sees the same tensors per waveform. **Primary classical comparison** is absolute R_{blend} against the non-AI board (no-bake, DualCosine, `seam_fir3`, poly, fades, VA residuals, Section 3.4). The search objective is maximize R toward the ideal sibling (best R=1). DualCosine is one classical comparator and a PPO centering constant. It is not the reward target.

Dataset statistics. Table 3 and Figures 4–5 inventory every evaluation board used in this paper. The primary procedural

Table 3: Dataset statistics (evaluation boards). L=256 unless noted. Hours are raw source duration on disk where measurable, not tiled period count. Plot 4 compares ≈1,280-height boards only (multi-family / Factory+FX / AKWF / transfer).

Board	Size	Notes
Holdout metrics	4,096 cycles	seed `20260719`, N=16 tiles
Score batch	64 cycles	classical / favorite tables
Multi-family SOTA	1,280 cycles	10 families × 2 seeds × 64
Factory+FX export	1,280 periods	640 dry + 640 FX mid-tile extracts
AKWF OA	1,280 periods	CC0 single-cycle WAVs
Meta hybrid_lstm	1,200 iters	$R_{\text{blend}}+J$, seed `1902771841`
CWRU	256 periods	4 MAT files, DE @12 kHz
MFPT	256 periods	shaft-rate windows
MIT-BIH	256 periods	10 records @360 Hz
PTB-XL subset	256 periods	94 `*_hr` @500 Hz
synth CNC / PMU	256+256	procedural pilots

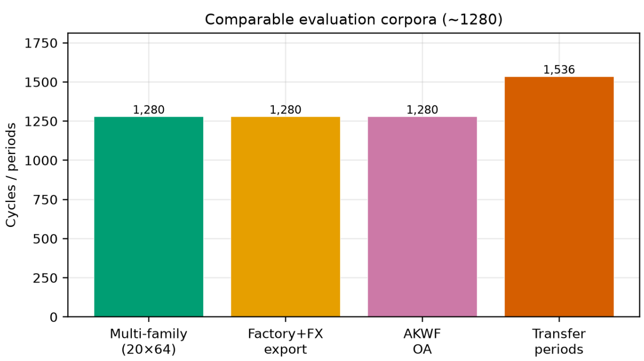

Figure 4: Corpus sizes across evaluation boards (cycles or periods). Holdout metrics draw, matched score batch, multi-family SOTA, factory/AKWF wavetable-native sets, and summed transfer periods.

holdout is a frozen 4,096-cycle metrics draw (L=256, N=16 tiles, seed `20260719`) plus a matched score batch of 64 cycles (five consecutive seeds for classical mean±std). Multi-family SOTA scores 20 waveforms (10 families × 2 seeds, batch 64, 1,280 cycles). Wavetable-native boards use ReelSynth Factory+FX exports (n=1,280: 640 dry + 640 FX) and AKWF CC0 cycles (n=1,280) so Plot 4 corpus bars are comparable to multi-family (1,280). Sci/eng transfer tiles 1,536 periods across CWRU, MFPT (n=256), MIT-BIH, PTB-XL subset, and synthetic CNC/PMU. Search never trains on the frozen holdout. N2N/seq baselines use disjoint train seeds `424242`/`424243`. Artifacts: dataset artifacts.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016, mean engine residual RMS 0.021 ± 0.013, and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634. No-bake (passthrough) on that draw scores mean $R \approx 0.971$. Method tables use a matched score batch of 64 cycles from the same seed (the canonical evaluation tensors), five consecutive seeds for mean±std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 6 summarizes the 4,096-cycle metrics draw: ideal RMS clusters near 0.72, wrap jumps span nearly three orders of magnitude (median ≈ 0.83), and no-bake R is high on easy tiles but falls below 0.91 on the hardest decile of wrap jumps. Figure 2 is the documented sine-wrap edge case: a high-$|wrap|$ holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

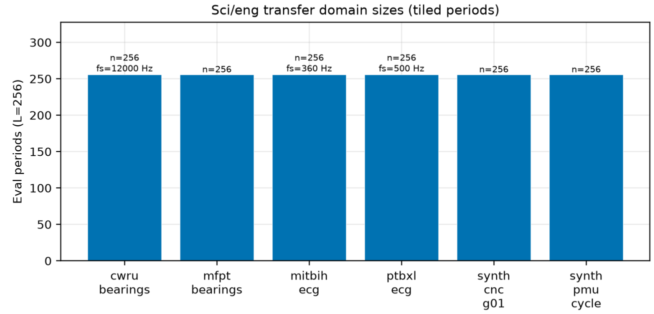

Figure 5: Sci/eng transfer domain sizes: tiled eval periods at L=256 (with source f_s where applicable).

Hybrid search campaign (5,000-iteration freeze). A CUDA search tagged `PPO+GA+PBT+NAS+depth+MoE` runs with a large declared iteration budget (order 10^5). At the reported freeze (run `gpu-rl-arch-20260719T083019Z`, ≈ 6,922 clean iters / ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Branch-best residuals at the freeze: GA ≈ 0.99016, PBT ≈ 0.99012, PPO ≈ 0.98924, combo ≈ 0.98854, NAS ≈ 0.98677. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 18. Figures in Section 5 regenerate from the archived search results.

Inference, classical, and learned baselines. High-R fitted operators ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 8). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are scored on the same holdout / cliff / multi-family protocol (Table 13 and the VA results) and illustrated on the intro high-wrap tile (Figure 12). Fixed-arch MLP-on-R and CNN/U-Net-lite-on-R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Phase B adds a primary corrupt→corrupt Noise2Noise

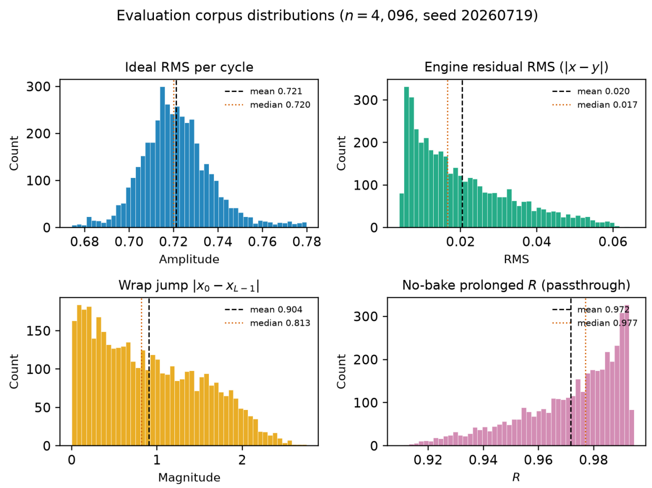

Figure 6: Holdout distribution statistics (seed `20260719`, n=4,096 cycles). Distinguish: ideal sibling $r^\star$ (cliff withheld), no-bake = unrepaired cracked engine (not $r^\star$), and DualCosine = one classical bake. Top left: ideal RMS per cycle. Top right: engine residual RMS before any bake. Bottom left: wrap jump magnitude $|x_0 - x_{L-1}|$ (cliff hardness proxy). Bottom right: no-bake prolonged R (passthrough, $x \mapsto x$). Dashed and dotted vertical lines mark mean and median. Hard-cliff reporting (Section 5) uses the top 10% wrap-jump stratum where no-bake R drops.

seam CNN (train seed `424242`, disjoint from holdout), a secondary sibling-supervised ceiling on the same architecture, and lightweight LSTM / depthwise 1D-CNN seq restorers (train seed `424243`). No search-champion weights initialize those baselines. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

Cliff strata and wavetable-native realism. A 4,096-tile holdout draw (seed `20260719`) is stratified by engine wrap-jump percentiles (top 25% / top 10%), with cutoffs in the cliff-strata data. Hard-cliff is the operative regime for claims about DualCosine collapse vs favorite retention. Edge RMSE is the locked required seam-local secondary. Click energy is optional. Wavetable-native transfer (Phase F1) scores true ReelSynth-exported factory periods (primary) and external AKWF CC0 WAVs (secondary) under the same close-seam→open-wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music LibriSpeech/MUSDB corpora remain off.

Audible demos and vibrato spectrograms. Downloadable clips (no-bake / DualCosine / Ours, 44.1 kHz mono, A4, 3 s): hear-sample WAVs. Rebuild script: `export_meta_hear_samples.py`. Slow-vibrato spectrogram protocol (`bench_vibrato_spectrogram.py`, artifacts: vibrato folder) scores the same bake operators under modulated-pitch playback. Figures appear in Section 5.17. These assets support informal audition only. Formal MOS/MUSHRA scores were not collected. The planned listening protocol is in Section 5.21.

4.2 Cross-domain wrap transfer protocol

The primary claim of this paper is wavetable seam repair. Transfer boards answer a narrower mechanistic question: does the *same* wrap-discontinuity protocol (open a cliff at a period join, score against a no-cliff ideal sibling, fit a compact repair operator) still make sense on other short periodic signals? They are stress tests of that geometry. They are *not* claims that an audio-trained bake transfers as bearing-fault or ECG diagnosis SOTA, and they are not claims that fatigue cracks or grid phase events "are" wavetable seams.

Shared transfer recipe. For each domain we cut fixed-length periods (L=256) and build an ideal sibling (smooth / template / cubic path) plus a cracked engine (same content with an open-wrap cliff, sometimes with a deliberately worse interpolant). We then score classical baselines and a domain-trained DenoiseOpt / Noise2Noise stack under locked R_{blend} (`hybrid_lstm` FitCell, 150–250 outer iterations where searched, `fit_steps` = 40). Classical baselines (no-bake, DualCosine, FIR, endpoint pin, ...) share that board. Domain-trained Noise2Noise (corrupt→corrupt, 4000 Adam steps) is scored on the same seven boards and reported beside classical rows in Table 15. Artifacts: signal_heal_transfer.

Seam analogy by domain family. Bearing revolutions (CWRU / MFPT / Paderborn). One shaft revolution is the period. The seam is the join between the last and first angle samples after resampling to L=256. A wrap cliff models a bad change-of-time / interpolant at that join, not a physics model of fatigue crack growth. Ideal siblings use a smoother angular path (cubic) than the cracked engine (linear + cliff).

ECG beats (MIT-BIH / PTB-XL subset). One heartbeat (R–R window) is the period. The seam is the join between beat end and beat start when the window is tiled. A wrap cliff is an artificial endpoint mismatch on that window. The ideal is a mild endpoint-equalized beat template. Clinical arrhythmia restore metrics are out of scope.

Synthetic CNC / PMU. Stand-ins for closed machine-tool contours and one AC / PMU voltage cycle when real KIT / IEEE DataPort downloads are walled. The seam is again a period-join cliff under the same residual protocol. These rows are procedural pilots, not industrial toolpath or live-grid claims.

What each dataset is (pointers). CWRU / MFPT / Paderborn K001: public vibration, n=256 revolutions each (Paderborn also hosts a separate Al Firdausi 4-class fault classifier, labeled apart from wrap-R_{blend}). MIT-BIH and PTB-XL 500 Hz lead-I subset: PhysioNet beat windows, n=256. Synthetic CNC G01 and synthetic PMU/AC: procedural, n=256.

Scope note. Transfer tables compare classical rows, domain-trained Noise2Noise, and Ours under this residual protocol. Author ECG Cycle-GAN and BeatDiff carry OOD

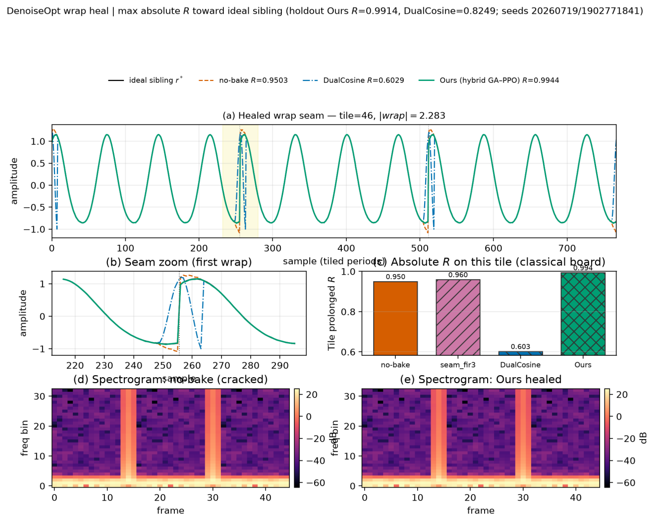

Figure 7: Holdout wrap-seam heal for the hybrid champion vs no-bake, Dual Cosine, and ideal sibling $r^\star$ (seed `20260719`).

wrap probes, while Paderborn Al Firdausi is a trained fault classifier (plus labeled arch-reuse wrap bake). Formal listening remains out of scope (Section 9 and Table 16). Results appear in Section 5.20.

5 Results

Reading guide. Primary numbers use locked R_{blend} (Sections 5.1–5.2). Hard-cliff classical collapse and multi-family ranking follow. Matched outer-loop bake-offs, HP probes, branch freezes, and transfer latency sit in Appendix B. Honesty notes are collected in Section 9.

5.1 Hybrid search under the blended score

Does hybrid GA+PPO beat the frozen Noise2Noise gate under R_{blend} and J? A 1,200-iteration search (seed `1902771841`, $\approx$1.65 h wall) selects on J (5) (λ=0.02). Champion: $R_{\text{blend}}\approx0.9695$, $J\approx0.9603$, $t\approx5.08$ ms. Search-monitor Dual Cosine on the same runner: $R_{\text{blend}}\approx0.504$. Frozen Noise2Noise holdout gate: $R_{\text{blend}}\approx0.9750$. The search champ does *not* clear that gate, while clearly clearing Dual Cosine. Artifacts: hybrid search data.

5.2 Noise2Noise vs Ours on four boards

Table 6 summarizes the four evaluation boards under R_{blend}, with Dual Cosine as the classical fade row on every board. Across five holdout seeds (`20260719`–`20260723`), Noise2Noise leads Ours on every seed (0.9767$\pm$0.0015 vs 0.9708$\pm$0.0014; Dual Cosine 0.542$\pm$0.038). On the 20-waveform multi-family board, Ours mean-of-means 0.9337$\pm$0.0028 edges N2N 0.9308$\pm$0.0045 (mean win fraction 0.47 across seeds; sign test on board-mean deltas p=1.0), and both dominate Dual Cosine. On balanced wavetable-native corpora (n=1,280 each), N2N leads Factory+FX (0.771 vs Ours 0.699, Dual Cosine 0.589) and AKWF (0.731 vs Ours 0.679, Dual Cosine 0.424).

Table 4: Parameter-count and inference-latency fairness under the locked holdout story. Favorite Θ is the v10.1 hybrid champion cell (search seed `1902771841`). SeamN2N is the frozen corrupt$\to$corrupt gate (train seed `424242`). Holdout R_{blend} is mean$\pm$std over five evaluation seeds. Latency is CUDA ms/batch at B=64 (warmup 8, timed 30; Table 5).

Method	Params	Holdout R_{blend}	ms/B=64
Favorite Θ (v10.1 hybrid)	$\approx$121k	0.9708$\pm$0.0014	8.20
SeamN2N (corrupt$\to$corrupt)	$\approx$53.5k	0.9767$\pm$0.0015	1.12
Dual Cosine	0	0.542$\pm$0.038	—

Table 5: Inference latency: Ours (hybrid favorite) vs Noise2Noise on CUDA (warmup 8, timed 30 synchronized forwards). Ratio = Ours/N2N ms per batch.

Board / batch	Ours ms	N2N ms	Ratio	Ours ms/samp
Holdout (B64)	8.20	1.12	7.36$\times$	0.128
Multi-family mean (B64)	7.50$\pm$0.56	1.06$\pm$0.07	7.12$\times$	0.117
Factory+FX (B64)	7.31	1.06	6.91$\times$	0.114
AKWF (B64)	7.50	1.11	6.74$\times$	0.117
Holdout (B1)	8.52	1.30	6.56$\times$	8.52
Holdout (B8)	7.71	1.07	7.21$\times$	0.964
Holdout (B256)	7.62	1.52	5.03$\times$	0.030

Table 4 states the size and latency comparison the review requested. The hybrid favorite is larger and slower than SeamN2N on this freeze ($\approx$7.4$\times$ holdout ms/batch). N2N still leads holdout R_{blend} on all five seeds. Search seeds, N2N train seed `424242`, and holdout seeds stay disjoint.

Multi-seed protocol. Primary holdout and multi-family rows in Table 6 report mean$\pm$std over five evaluation seeds. Matched-5k outer-loop re-search under $R_{\text{blend}}+J$ across three search seeds is complete (`meta_approach_compare_v13_rblend/`; Table 17: Ours 0.9748$\pm$0.0018, Aging 0.9635$\pm$0.0049).

5.3 Ours vs Noise2Noise inference latency

Table 5 compares forward-pass wall time on CUDA across boards and batch sizes. At the paper-default batch B=64, Ours is $\approx$7$\times$ slower than N2N on every board (holdout 8.20 vs 1.12 ms; multi-family mean 7.50$\pm$0.56 vs 1.06$\pm$0.07 ms; Factory+FX 7.31 vs 1.06 ms; AKWF 7.50 vs 1.11 ms). The holdout batch sweep ($B\in\{1,8,64,256\}$) keeps the ratio in $\approx$5–7$\times$: larger batches amortize per-sample cost for both methods, but N2N remains faster. This matches the parameter gap ($\approx$121k vs $\approx$53.5k) and the J latency penalty in (5). Artifacts: `v13_ours_vs_n2n_latency/latency_compare.json`.

5.4 Search efficiency (learning curves)

Champion score versus outer-loop evaluation index is plotted in Figure 8. The left panel compares a matched 5,000-evaluation hybrid to Random NAS under *superseded* whole-curve prolonged R (seed `1902771841`, Dual Cosine $\approx$0.817), where both arms clear Dual Cosine early. The right

Table 6: Ours vs Noise2Noise and Dual Cosine under R_{blend} (α=0.7). Holdout and multi-family rows: mean±std over five evaluation seeds (`20260719`–`20260723`). Factory+FX / AKWF remain single-corpus freezes. Δ columns are Ours minus that baseline.

Board	Ours	N2N	DualCosine	Δ_{N2N}	Δ_{DC}	Gate
Holdout	0.971±0.001	0.977±0.001	0.542±0.038	−0.006	+0.429	0/5
Multi-family (mean)	0.934±0.003	0.931±0.005	—	+0.003	—	0.47
Factory+FX	0.6990	0.7714	0.5887	−0.0724	+0.1104	no
AKWF OA	0.6794	0.7305	0.4244	−0.0511	+0.2551	no

panel is the locked R_{blend} / J hybrid run over 1,200 evaluations (seed `1902771841`): raw champion R_{blend} clears Dual Cosine immediately and plateaus below the Noise2Noise gate (≈0.977). Matched-5k multi-seed re-search under R_{blend} is complete (Figure 22; Table 17). Plateau boredom retunes branch mix (Section 3.7). Near-ceiling residuals climb only a little after the first few hundred evaluations. Champion parameter count is not fixed: hybrid live champs are ≈24k±2k params across seeds, while mid-search updates swing from ≈15k to ≈370k.

Superseded whole-curve freezes. Whole-curve prolonged-R freezes (R≈0.991 vs DualCosine ≈0.825) are not current results. Selection and published comparison under v10.1 use $R_{\text{blend}}+J$ with Noise2Noise as the primary neural baseline.

5.5 Which families stay hard

On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Superseded 5k-gate freeze. At the reported 5k-gate search freeze (≈ 6,922 clean iterations, ≈ 7.9 h on RTX 3090) the champion residual was $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the CUDA search runner ($\Delta R \approx +0.174$). That freeze optimized whole-curve prolonged R. It is superseded by R_{blend} for v10.1 selection. Monitoring plots below come from that older freeze and should be read as search-trace artifacts, not as locked-protocol scores.

5.6 Inference latency at matched score

Across the prior high-prolonged-R fitted operators, we retimed CUDA forward passes under locked R_{blend} (batch 64). Live R_{blend} spreads more than the old prolonged band: the max-R_{blend} model is `champion_iter_000265` (R_{blend}≈0.9769, ≈ 6.53 ms/batch, ≈ 94k params). The compact historically favorite `iter_000235` remains lowest-latency among the top-5 distinct topologies (R_{blend}≈0.9749, ≈ 3.42 ms/batch, ≈ 37k params). Selection rule: maximize live R_{blend}, then minimize latency inside $R \geq R_{\max} - 5 \cdot 10^{-4}$.

Table 7: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64) under R_{blend} (α=0.7). R_{saved} is the checkpoint hint (often prolonged R); R_{live} is measured R_{blend}. *Max-R_{blend} (also sole member of the $R \geq R_{\max} - 5 \cdot 10^{-4}$ band here).

Tag	R_{saved}	R_{live}	ms/batch	params
`iter_000265`*	0.9906	0.9769	6.53	93 566
`iter_000140`	0.9906	0.9762	7.16	104 484
`iter_000133`	0.9908	0.9752	7.16	104 484
`iter_000229`	0.9909	0.9751	7.56	100 500
`iter_000235`	0.9910	0.9749	3.42	37 489

*Favorite under maximize-R_{blend} then latency. Compact `iter_000235` is still the latency floor in this top-5.

Table 8: Canonical holdout under R_{blend} (α=0.7; seed `20260719`). Mean±std over five seeds. No-bake = unrepaired engine. Favorite = locked v10.1 hybrid champ.

Method	R_{blend}	R_{blend} (5-seed)	ms/batch
neural favorite	0.9697	0.9708 ± 0.0012	5.99
no-bake (passthrough)	0.9142	0.9166 ± 0.0090	1.06
`seam_fir3`	0.8842	0.8879 ± 0.0087	1.11
classic quadratic	0.5742	0.5727 ± 0.0318	2.07
DualCosine	0.5409	0.5420 ± 0.0342	1.66
cosine fade	0.5409	0.5420 ± 0.0342	1.87
crossfade	0.5310	0.5273 ± 0.0311	1.86
hann blend	0.5197	0.5214 ± 0.0337	1.05
soft wide fade	0.4828	0.4803 ± 0.0258	2.85
detrend	0.3437	0.3453 ± 0.0399	0.96
ensemble detrend+DC+FIR	0.3126	0.3100 ± 0.0421	2.00

5.7 Top-5 architectures

Table 7 ranks five distinct fitted bake graphs from the CUDA inference bench (inference data, batch 64), preferring live R_{blend} first and latency on ties, and skipping near-duplicate copies of the same topology signature.

5.8 Classical methods vs neural favorite

Table 8 scores the classical (non-AI) bake set (no-bake, DualCosine, and `seam_fir3`) plus the locked v10.1 hybrid favorite on the frozen corpus of Section 4.1 (CUDA, batch 64, seed `20260719`) under R_{blend}. DualCosine reaches $R_{\text{blend}} \approx 0.541$ (≈ 1.66 ms/batch). Active classical `seam_fir3` reaches 0.884 / ≈ 1.11 ms/batch (0.888 ± 0.009 over five seeds). No-bake (passthrough) scores 0.914 (0.917 ± 0.009). The hybrid favorite reaches 0.970 (0.971 ± 0.001 over five seeds) at ≈ 5.99 ms/batch: $\Delta R_{\text{blend}} \approx +0.056$ vs no-bake and ≈ +0.429 vs DualCosine. Noise2Noise on the multi-family matrix is Table 9. The four-board gate is Table 6.

5.9 Multi-family comparison

Does the locked hybrid favorite beat the classical non-AI board (no-bake, DualCosine, FIR, fades), fixed MLP/CNN baselines, *and* Noise2Noise across twenty generative families

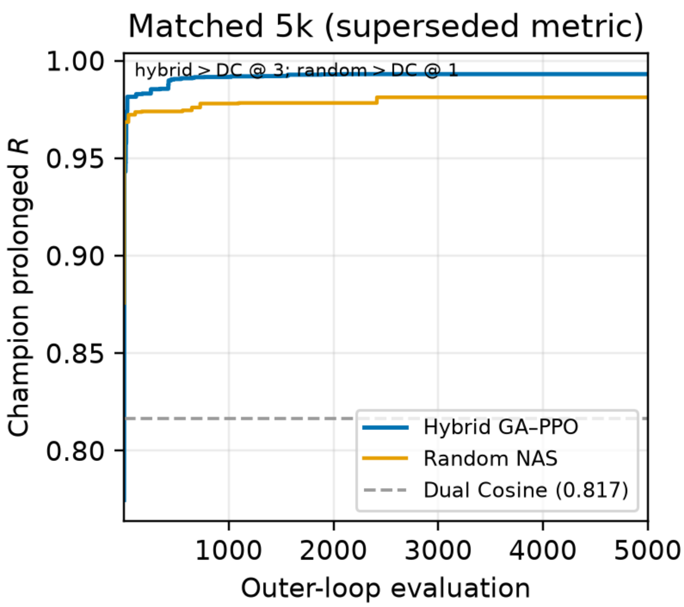

Figure 8: Search learning curves (seed `1902771841`). Left: champion prolonged R for hybrid GA–PPO vs Random NAS on the matched 5k bake-off (**superseded** whole-curve metric; Dual Cosine dashed). Right: v10.1 hybrid champion J and raw R_{blend} over 1,200 evaluations (Dual Cosine and Noise2Noise gate marked; matched-5k multi-seed R_{blend} bake-off in Appendix B).

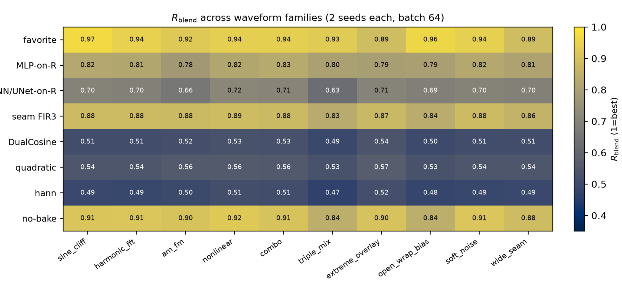

Figure 9: Colorblind-safe (`cividis`) heatmap of mean R_{blend} for selected methods × generative families (2 seeds each). Higher is better.

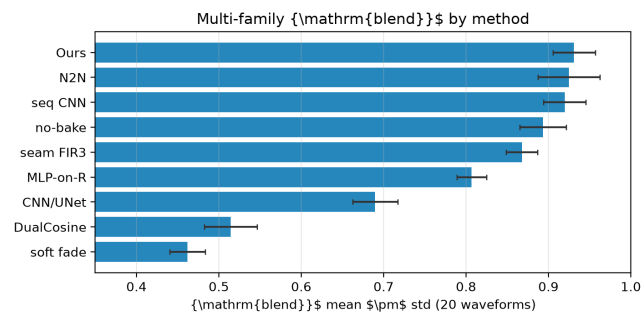

Figure 10: Multi-family mean R_{blend} by method (same 20-waveform board as Table 9).

under R_{blend}? Table 9 reports mean±std over the 20-waveform multi-family set (Section 4.1) from the SOTA matrix. Figures 9–10 plot the same ranking. Primary reading is absolute R_{blend} rank: Ours 0.934, corrupt→corrupt N2N 0.931, seq CNN 0.920, no-bake 0.893, `seam_fir3` 0.868, DualCosine 0.514 (aligned with Table 6 multifamily). Fixed MLP-on-R and CNN/U-Net-lite-on-R (no outer NAS) sit between DualCosine and the neural rows. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: $W{=}0$, $p{\approx}8.9{\times}10^{-5}$, mean $\Delta R_{\text{blend}}{=}{+}0.417$ (bootstrap 95% CI $[0.398, 0.435]$). Vs no-bake the multifamily mean gap is $\approx +0.038$. Vs corrupt→corrupt N2N the mean-of-means gap is $\approx +0.003$ (mean win fraction 0.47 across five seeds).

5.10 Hard cliffs expose classical fade failure

Does high no-bake R_{blend} hide DualCosine failure on large wrap jumps? Table 10 stratifies a 4,096-tile holdout draw (seed `20260719`) by engine wrap-jump percentiles (p75≈ 1.387, p90≈ 1.832), frozen in the cliff-strata data under R_{blend}. No-bake R_{blend} can stay high on mild cliffs because residual mass remains small relative to ideal RMS. DualCosine collapses on hard cliffs (0.525 → 0.305), while the hybrid favorite holds (0.971 → 0.978) and corrupt→corrupt N2N stays slightly ahead (0.976 / 0.983 on all / top-10%). Favorite wrap-jump on the top-10% subset stays ≈ 1.82 (vs no-bake 2.09). The claim is tiled residual repair, not endpoint-zeroing. **Edge RMSE** is the locked required seam-local secondary (EVAL_PROTOCOL): on the top-10% subset it drops from no-bake 0.095 to favorite 0.027 (N2N corrupt→corrupt 0.020, DualCosine 0.990). Click energy remains an optional diagnostic.

5.11 Noise2Noise and sequence baselines

How does DenoiseOpt compare to true Noise2Noise-style and lightweight seq restorers on the same generator under R_{blend}?

Table 9: Multi-family board (20 waveforms) under R_{blend} (α=0.7). ΔR^* and ms* vs Dual Cosine only. N2N/seq use train seeds `424242`/`424243`.

Method	R_{blend} (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	ΔR^*
neural favorite	0.931 ± 0.026	34.6 ± 3.2	34.7 ± 3.2	0.89 ± 0.14	4.50	121 451	+0.417
N2N corrupt→corrupt	0.925 ± 0.038	33.8 ± 4.0	33.9 ± 4.0	0.89 ± 0.15	0.74	53 506	+0.410
N2N sibling-sup.	0.924 ± 0.035	33.9 ± 4.1	34.0 ± 4.1	0.91 ± 0.15	0.74	53 506	+0.410
seq CNN1D	0.920 ± 0.026	32.2 ± 3.5	32.4 ± 3.5	0.92 ± 0.15	0.53	3 242	+0.405
no-bake	0.893 ± 0.028	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.379
seq LSTM	0.891 ± 0.075	32.2 ± 5.8	32.4 ± 5.8	0.86 ± 0.15	1.53	75 746	+0.376
`seam_fir3`	0.868 ± 0.019	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.17	0	+0.353
MLP-on-R	0.807 ± 0.018	24.7 ± 0.9	24.8 ± 0.9	0.59 ± 0.07	1.76	8 604	+0.292
CNN/U-Net-on-R	0.690 ± 0.027	19.5 ± 0.9	19.5 ± 0.9	0.47 ± 0.06	3.08	11 752	+0.175
classic quadratic	0.547 ± 0.031	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.02	0	+0.032
DualCosine	0.514 ± 0.032	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	0.98	0	0
cosine fade	0.514 ± 0.032	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.07	0	0
crossfade	0.505 ± 0.032	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	0.95	0	−0.010
hann blend	0.495 ± 0.032	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.19	0	−0.020
soft wide fade	0.462 ± 0.021	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	1.87	0	−0.053
detrend	0.311 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.06	0	−0.203
ensemble DC+FIR	0.275 ± 0.043	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.05	0	−0.239

Table 10: Cliff strata on 4,096 holdout tiles under R_{blend} (α=0.7). Strata by engine wrap-jump.

Method	all R_{blend}	all jump	top25 R_{blend}	top25 jump	top10 R_{blend}	top10 jump
seq LSTM (ceiling)	0.986 ± 0.009	0.865	0.989	1.637	0.990	1.810
N2N sibling-sup.	0.981 ± 0.013	0.874	0.985	1.656	0.985	1.827
N2N corrupt→corrupt	0.976 ± 0.015	0.861	0.981	1.631	0.983	1.804
neural favorite	0.971 ± 0.020	0.863	0.978	1.634	0.978	1.817
seq CNN1D	0.960 ± 0.028	0.881	0.969	1.684	0.970	1.871
no-bake	0.913 ± 0.077	0.913	0.929	1.797	0.922	2.094
`seam_fir3`	0.886 ± 0.067	0.456	0.870	0.893	0.869	1.034
DualCosine	0.525 ± 0.239	0.335	0.318	0.290	0.305	0.335

n: all 4096, top25 1024, top10 410. $\pm$ is tile std on the all stratum.

Table 11: Required seam-local secondary on top-10% wrap-jump tiles (n=410): edge RMSE (lower better). Frozen in the cliff-strata data.

Method	top10 edge RMSE
seq LSTM	0.011
N2N sibling-sup.	0.017
N2N corrupt→corrupt	0.020
neural favorite	0.027
seq CNN1D	0.035
no-bake	0.095
`seam_fir3`	0.164
DualCosine	0.990

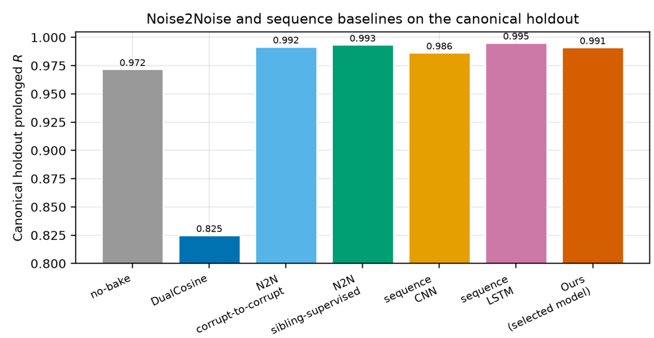

Figure 11: Canonical holdout R_{blend}: N2N and seq baselines vs Ours (locked hybrid). Train seeds `424242`/`424243` are disjoint from holdout `20260719`.

Primary corrupt→corrupt N2N (train seed `424242`) reaches holdout $R_{\text{blend}} \approx 0.975$ (Table 6), above the hybrid favorite (0.970). On the 4,096-tile cliff draw, sibling-supervised N2N reaches 0.981 and seq LSTM 0.986 (Table 10). Holdout tiles never enter training. Classical DualCosine / `seam_fir3` remain in Table 10. Figure 11 plots the four-board R_{blend} comparison.

Table 12: Wavetable-native wrap protocol under R_{blend} (α=0.7). Close-seam ideal → open-wrap cliff (seed `20260719`). Primary = ReelSynth export. Secondary = AKWF CC0 files.

Method	Export	AKWF OA
`seam_fir3`	0.863	0.725
poly seam (d=3)	0.846	0.696
no-bake	0.845	0.702
endpoint-pin (mean)	0.792	0.745
N2N corrupt→corrupt	0.771	0.731
seq CNN1D	0.706	0.677
neural favorite	0.699	0.679
DualCosine	0.589	0.424
seq LSTM	0.628	0.579

5.12 Factory exports and AKWF cycles

Does the wrap protocol transfer beyond synthetic sine cliffs onto true exports and third-party OA cycles? Table 12 scores ReelSynth-exported factory periods (primary realism) and Adventure Kid Waveforms (AKWF) CC0 single-cycle WAVs (secondary) after close-seam idealization and the same open-wrap cliff, under R_{blend}. Procedural stand-ins are demoted to tertiary smoke and no longer carry the primary realism claim. On exported factory periods the favorite (R_{blend}≈0.699) trails no-bake / FIR / poly / N2N while still beating Dual Cosine (0.589). On AKWF OA cycles the favorite (0.679) trails no-bake / N2N and beats Dual Cosine (0.424). Export gaps are listed in Section 9.

5.13 Polynomial baseline and endpoint pinning

Does a cheap classical poly fitter close the residual, and does endpoint pinning show the wrap-jump vs R_{blend} tradeoff? A seam-local degree-3 polynomial fitter (no learning) reaches canonical $R_{\text{blend}} \approx 0.917$ (slightly above no-bake 0.914) with wrap-jump essentially unchanged (polynomial results). SSM seq models are deferred. The LSTM remains the supervised seq ceiling. An endpoint-pin control (pinning results) drives wrap-jump to 0 on all / hard subsets while dropping R_{blend} (all 0.733, hard 0.597). Methods that zero wrap-jump are not the same objective as R_{blend} (Limitations).

5.14 Classical virtual-analog seam repairs

Do the cited VA antialias tools [1–3] beat DualCosine on R_{blend} when applied as cycle-local seam residuals?

What each operator changes. Figure 12 shows the family on one high-wrap sine+cliff tile (same seed and tile as Figure 2). **BLIT/BLEP** (Stilson/Smith lineage) treats the wrap as a value step. It subtracts a band-limited step residual (here a raised-cosine BLEP-family lobe scaled by the measured wrap jump) so the tiled click spectrum is softened near the seam. **PolyBLEP** (Nam et al.) is a cheap polynomial surrogate of that step residual with fractional-delay support near phase wrap. It edits only a few samples on each side of the discontinuity. **BLAMP** (Esqueda et al.) corrects a slope or corner jump (discontinuity in the first derivative), which matters for triangles, rectification, and hard clipping. On a pure value cliff the slope jump is small, so BLAMP is nearly a no-bake passthrough. **DualCosine** (our classical bake baseline) fades both ends of the stored period with raised-cosine windows. It does not insert a BLEP-style residual. It trades wrap energy for a wide end-fade distortion that R_{blend} penalizes on hard cliffs.

Scores. Table 13 reports batch means from the VA results (seed `20260719`, batch 64) under R_{blend}. Code is in the ReelSynth repository: PolyBLEP matches `osc::va::poly_blep`, BLIT/BLEP uses a raised-cosine step residual, and BLAMP applies a polyBLAMP lobe to the wrap slope jump. PolyBLEP reaches mean $R_{\text{blend}} \approx 0.795$ ($\Delta \approx +0.254$ vs DualCosine) but trails no-bake / `seam_fir3`. BLIT/BLEP nearly zeros wrap-jump (0.004), yet mean R_{blend} falls to 0.632 and hard cliffs collapse to 0.351. BLAMP stays near no-bake on this value-cliff geometry (mean 0.914). On the annotated severe tile in Figure 12, tile R_{blend} is lower still (PolyBLEP ≈ 0.726, BLIT/BLEP ≈ 0.385, Dual Cosine 0.300, BLAMP ≈ engine). We score these classical tools, not only cite them: they beat DualCosine on mean R_{blend} in places, and none matches the locked hybrid favorite on holdout R_{blend}.

5.15 When search helps vs classical methods

Per-cycle ΔR on export, AKWF, Rust `sound_bench`, and sine+cliff holdout (transfer results): favorite win-rate vs Dual Cosine is high on sine cliffs (1.00) and AKWF (0.88), and moderate on export (0.64) / Rust (0.75). Win-rate vs no-bake is high only on sine cliffs (0.92) and collapses on export (0.24) / Rust (0.20). Search value is highest on hard cliffs and when compact bake graphs are needed without engine→ideal labels at search time. Export / Rust gaps vs no-bake are listed in Section 9.

5.16 Transfer to Rust engine tiles

Table 14 scores the Python-trained favorite on 20 Rust-exported tiles under R_{blend}. The favorite beats Dual Cosine on mean ΔR_{blend}+0.082, but the signed-rank test is not significant at α=0.05 (p≈0.30), and absolute score trails no-bake and `seam_fir3`. The primary SOTA claim remains the Python multi-family matrix.

5.17 Vibrato spectrograms

Do wrap-seam differences stay visible under slow vibrato, beyond static tiled periods? Figure 17 renders the same holdout cracked period, DualCosine bake, and refit **Ours (hybrid GA–PPO)** champion under shared sinusoidal vibrato (±3% depth at 5 Hz around 220 Hz). Panels (a–c) show magnitude spectrograms. Panels (d–e) show DualCosine−no-bake and Ours−no-bake differences. Panel (g) reports cycle-local prolonged R alongside a modulated-playback residual under the same pitch envelope. On the annotated high-wrap tile, Ours retains high cycle R and high modulated R, while DualCosine remains the weakest classical

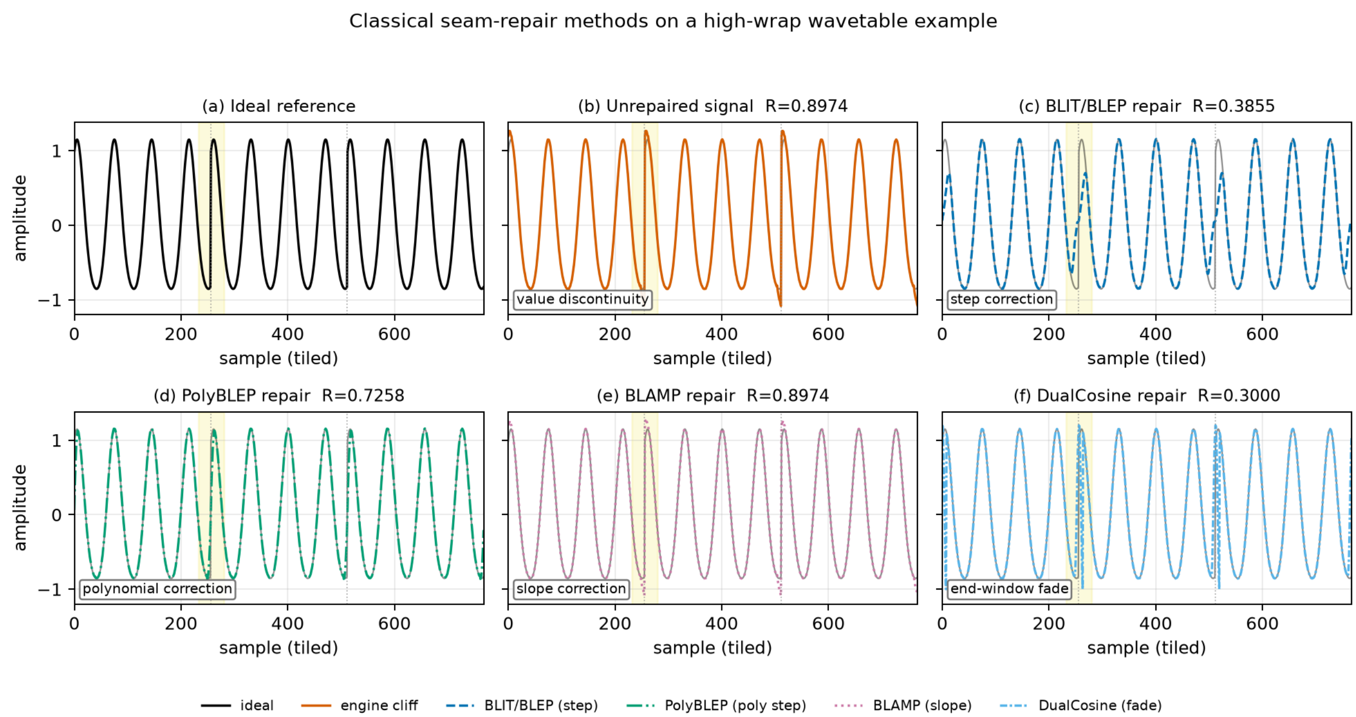

Figure 12: Classical VA seam techniques on one cracked sine+cliff tile (seed `20260719`, tile 46, $|wrap|$=2.283, L=256, three tiled periods). Yellow band marks the first seam neighborhood. Panel tile R_{blend} (1 = best): (a) ideal (unscored sibling). (b) cracked engine 0.897. (c) BLIT/BLEP 0.385. (d) PolyBLEP 0.726. (e) BLAMP 0.897 (near no-bake on value cliffs). (f) DualCosine 0.300. Colorblind-safe hues with distinct linestyles for B&W print. Batch means appear in Table 13. Accessibility: six-panel comparison of ideal, cracked, BLIT/BLEP, PolyBLEP, BLAMP, and DualCosine.

Table 13: Classical VA seam repairs under R_{blend} (α=0.7; seed `20260719`). Canonical holdout: R_{blend}, SNR/SDR, wrap-jump, edge RMSE. Cliff top-10%: R_{blend} (n=410). Multi: mean R_{blend} over 20 generative waveforms.

Method	R_{blend}	SNR	SDR	jump	edge	top10 R_{blend}	multi R_{blend}
BLAMP	0.9142	32.6	32.7	0.873	0.080	0.9219	0.893
no-bake	0.9142	32.6	32.7	0.873	0.080	0.9220	0.893
`seam_fir3`	0.8842	29.6	29.6	0.438	0.112	0.8693	0.868
PolyBLEP	0.7949	25.9	25.8	0.102	0.205	0.6895	0.784
BLIT/BLEP	0.6324	20.4	20.3	0.004	0.365	0.3510	0.612
DualCosine	0.5409	18.0	17.8	0.292	0.506	0.3049	0.514

Latency (canonical, CUDA): PolyBLEP ≈ 0.87 ms/batch, DualCosine ≈ 1.46, BLIT/BLEP ≈ 71, BLAMP ≈ 20. Frozen in the VA results.

Table 14: Rust `sound_bench` transfer (n=20) under R_{blend} (α=0.7). ΔR_{blend} vs Dual Cosine.

Method	R_{blend} (20 Rust tiles)	ΔR_{blend} vs Dual Cosine
no-bake	0.846 ± 0.171	+0.422
`seam_fir3`	0.751 ± 0.157	+0.327
neural favorite	0.506 ± 0.236	+0.082
classic quadratic	0.474 ± 0.186	+0.050
Dual Cosine / cosine fade	0.424 ± 0.187	0

comparator. This is objective spectrogram / residual evidence only. It is not a formal listening study. Protocol: `bench_vibrato_spectrogram.py`. Artifacts: vibrato folder.

5.18 Downloadable audio samples

As an audible companion to Figure 7, we release five fixed-pitch (A4, 3 s, 44.1 kHz mono) wavetable clips for no-bake, DualCosine, and Ours-healed periods from the matched-suite holdout (hear-sample WAVs, rebuild: `export_meta_hear_samples.py`). Figure 18 shows short waveform excerpts from those WAVs with per-clip absolute R. Readers can download and listen. We do *not* report formal MOS/MUSHRA or claimed A/B listening scores.

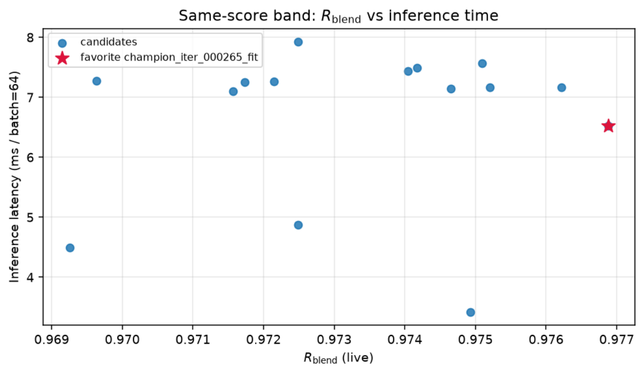

Figure 13: High-R fitted operators: residual vs CUDA latency. Star: favorite.

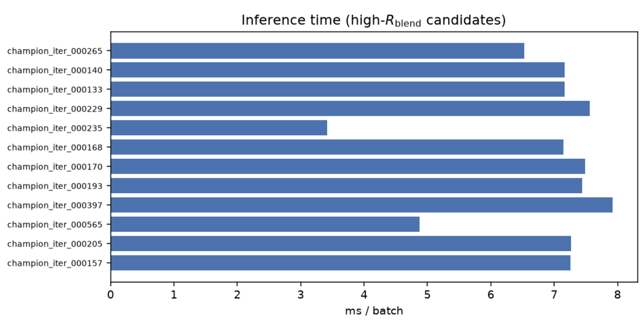

Figure 14: Inference latency for high-R fitted models (CUDA, batch 64).

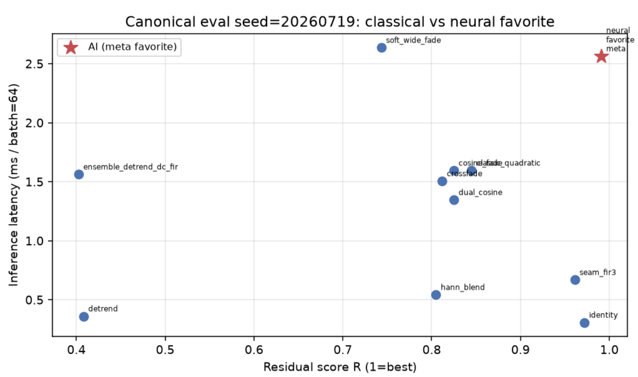

Figure 15: Canonical holdout: classical methods and favorite in the R vs ms/batch plane.

5.19 Factory and Adventure Kid waveform examples

Beyond the synthetic sine+cliff narrative, Figure 19 shows a compact gallery of true ReelSynth-exported factory periods (one mid-morph frame per bank) and Adventure Kid Waveforms (AKWF) CC0 single-cycle files already used by the wrap protocol. Scored means remain in Table 12 and the wavetable results. This panel is diversity evidence only. No new NAS was run for the gallery.

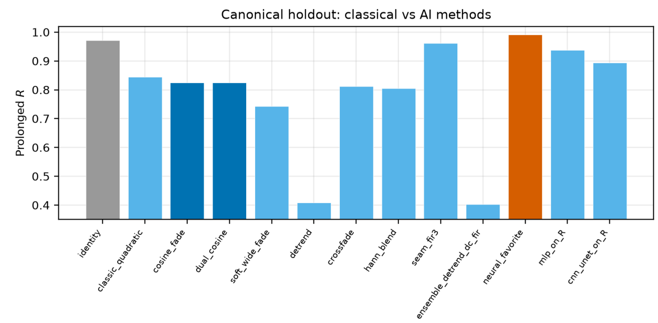

Figure 16: Canonical holdout ranking on R_{blend} (left) and latency (right).

5.20 Transfer to bearings, ECG, and synthetic cycles

Setup reminder. Table 15 reports locked R_{blend} after training/searching DenoiseOpt (`hybrid_lstm`, seed `1902771841`) on each transfer board from Section 4.2 (n=256 periods, L=256). Columns are: CWRU/MFPT/Paderborn bearing revolutions, MIT-BIH/PTB-XL ECG beats, and synthetic CNC/PMU pilots. Numbers come from the released results_table.json (`primary_metric=r_blend`). Domain-trained Noise2Noise-style SeamN2N (corrupt→corrupt, 4000 Adam steps, train seed `424242`, holdout seed `20260719`) is scored on the same seven boards. Author ECG CycleGAN and BeatDiff remain OOD wrap probes on ECG only (clinical restore ≠ wrap residual, Table 16). Artifacts: signal_heal_transfer.

Takeaways. Under R_{blend}, Ours leads every classical-board column in Table 15, including Dual Cosine and SeamN2N on every domain. SeamN2N beats Dual Cosine everywhere and sits near no-bake on several bearing / ECG columns, but trails Ours on ECG and synth CNC. On MIT-BIH and PTB-XL, endpoint-pin remains a strong classical ECG row above SeamN2N. Ours still leads both. Author Operational CycleGAN and BeatDiff collapse under OOD wrap scoring relative to Ours / SeamN2N / no-bake. We do not treat clinical morph metrics as wrap residual. Synthetic CNC shows the largest classical collapse with Ours near 0.991. Synth PMU is the strongest SeamN2N row (0.9589) but still below Ours (0.9919). Paderborn wrap board shows Ours 0.9369 vs SeamN2N 0.8421 and DualCosine 0.4710. Separately, we train Al Firdausi CNN as a 4-class fault classifier and do not rename classification as wrap residual (Table 16). Latency timings in Appendix B remain wavetable-checkpoint zero-shot. Primary claim remains cycle-local wavetable seam restoration.

5.21 Listening protocol (no formal listening study)

Downloadable WAVs (hear-sample WAVs) and vibrato spectrograms support informal A/B inspection (Sections 5.17–5.18). Formal MOS / MUSHRA scores were not collected. Until a formal study runs, perceptual claims remain qualitative and secondary to residual scores. See Section 9.

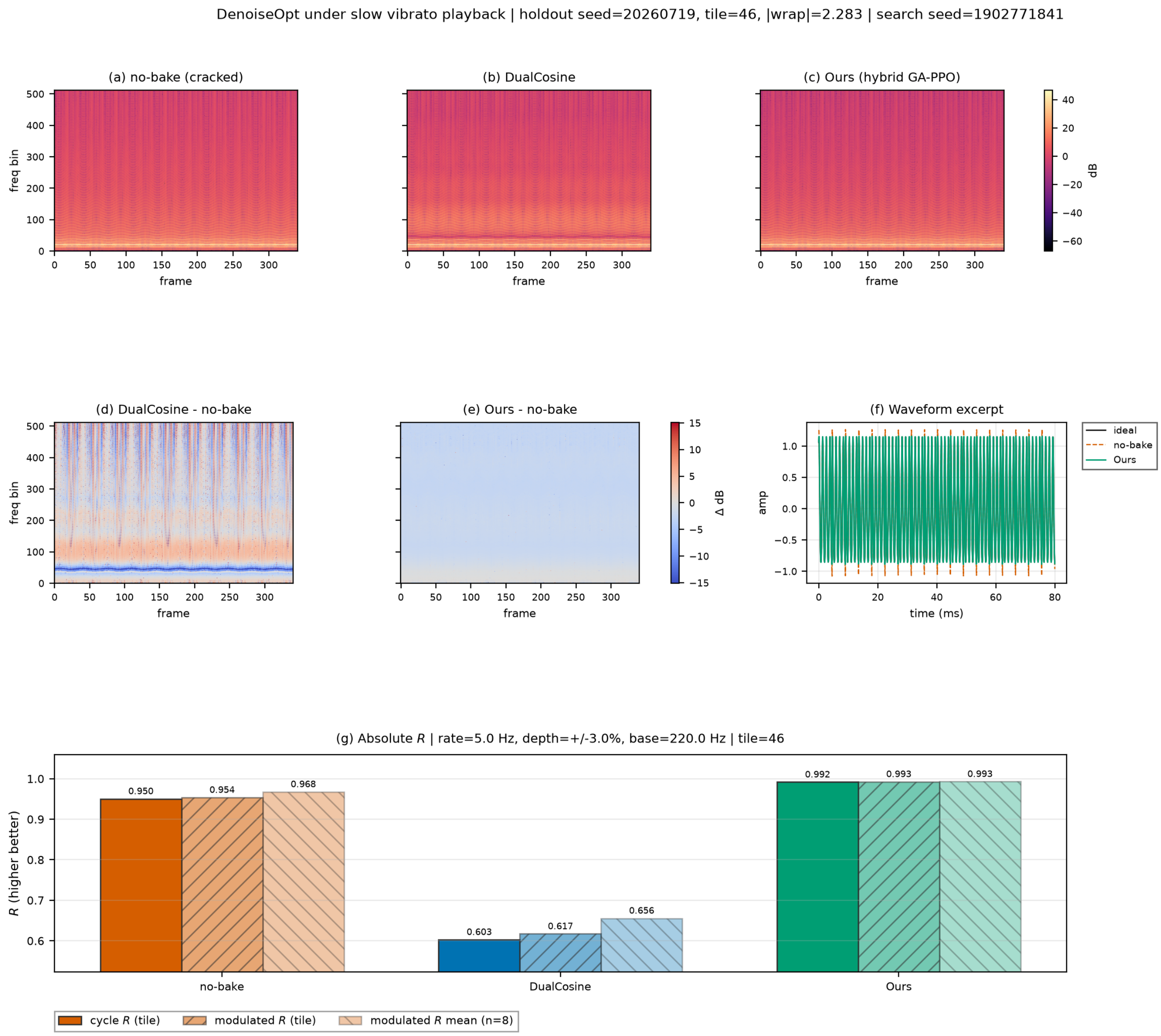

Figure 17: Slow-vibrato spectrogram eval on a high-wrap holdout tile (seed `20260719`). (a–c) Magnitude spectrograms for no-bake, DualCosine, and Ours. (d–e) Spectrogram differences vs no-bake. (f) Short waveform excerpt. (g) Absolute R: cycle-local prolonged residual, modulated-playback residual on the same tile, and mean modulated R over top-wrap tiles. Accessibility: Okabe–Ito hues. Difference maps use diverging coolwarm. Formal human A/B scores are out of scope.

6 Discussion

Main story under R_{blend}. The locked protocol does not crown the hybrid champion over Noise2Noise on holdout (0.9708±0.0014 vs 0.9767±0.0015 over five seeds). Dual Cosine on that board sits near 0.542±0.038. The operative claim is narrower. Compact searchable repair operators outperform classical fades, stay competitive with Noise2Noise on multi-family means (mean win fraction 0.47), and hold on hard cliffs where Dual Cosine collapses (Table 6, Section 5.10). Favorite Θ on the v10.1 freeze is larger than SeamN2N (≈121k vs ≈53.5k) and still trails N2N on holdout R_{blend} (Table 4). Whole-curve numbers near R≈0.991 come from superseded freezes. They are not current results under the locked protocol (Limitations).

Which families stay hard. Across large procedural benches, wrap error is uneven. Families such as `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` dominate the residual budget. `harmonic_fft` and `am_fm` are easier. Mean scores hide that structure.

Search near the ceiling. Under near-ceiling residuals, further gains are small in absolute units and sensitive to re-

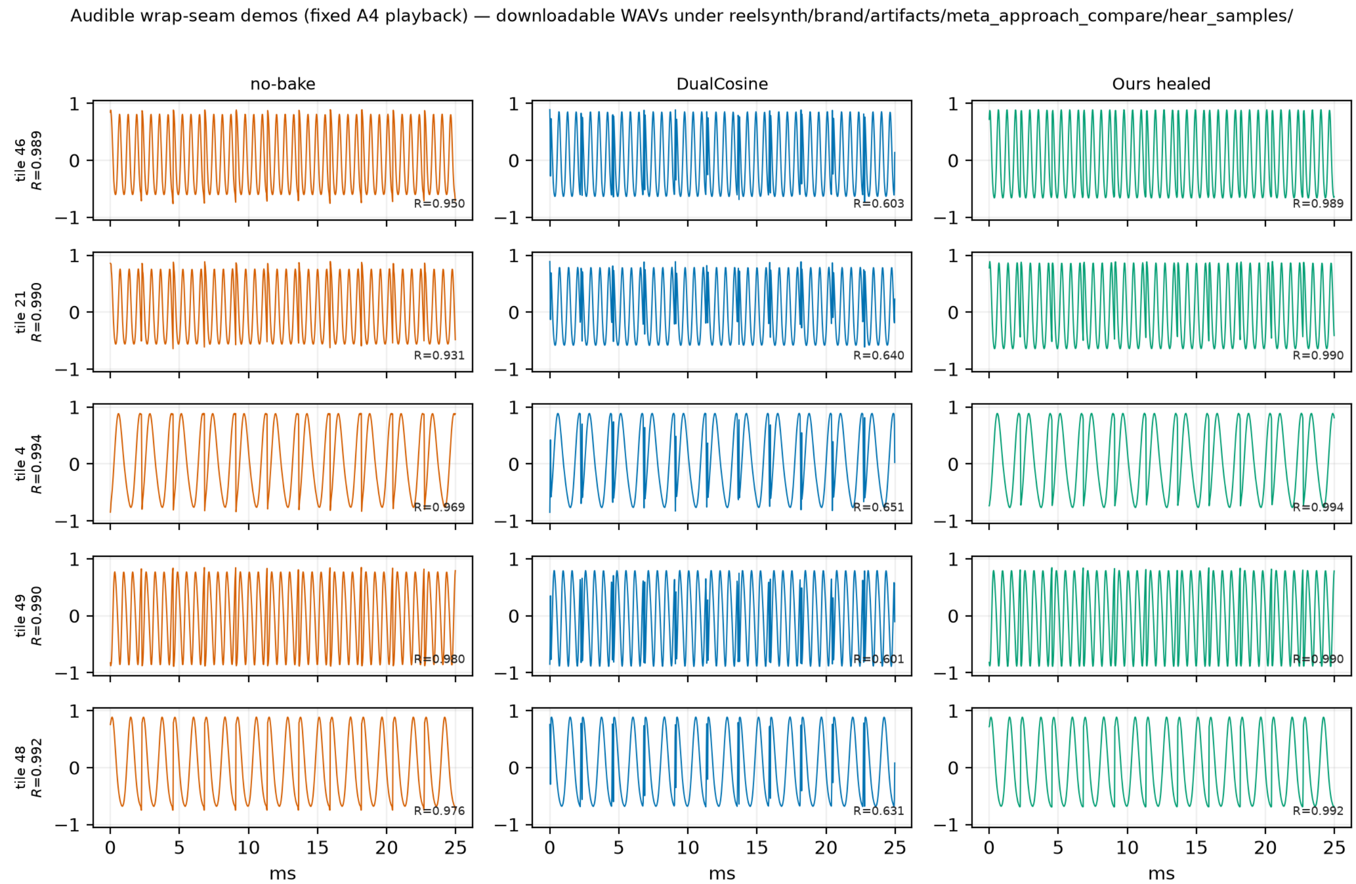

Figure 18: Hear-sample panel from the online WAV pack (hear-sample WAVs, five holdout tiles, no-bake / DualCosine / Ours). Waveforms show the first ≈ 25 ms of each clip. R annotations are cycle-local prolonged residuals vs the ideal sibling. Not a formal listening study.

ward shaping and branch weights. Learning curves (Figure 8) show Dual Cosine cleared early under both the superseded prolonged-R 5k bake-off and the v10.1 R_{blend} hybrid run, with a later plateau below the Noise2Noise gate. Plateau adaptation (deeper operators, denser mixture-of-experts graphs, higher GA/PBT weight) is the operational response inside one campaign. Matched-5k outer-loop bake-offs under locked $R_{\text{blend}}+J$ are complete across three search seeds (Appendix B, Table 17).

Evidence beyond static tiles. Vibrato spectrograms and downloadable hear WAVs support informal inspection (Sections 5.17–5.18). Sci/eng wrap transfer is a wrap-protocol stress test with per-domain seam analogies (Section 5.20). Perceptual and scope limits are collected in Section 9.

7 Tooling and AI use

AI-assisted workflow. The following tools assisted preparation. The authors checked all claims, numbers, and citations against code and frozen JSON artifacts (manual evidence alignment, not formal verification).

- **Cursor** assisted software development (search runners, baselines, transfer benches, and figure scripts in the ReelSynth repository).
- **SciSpace** assisted literature discovery and paper triage.
- **NotebookLM** assisted citation checks (bibliography entries cross-checked against open-access sources and the manuscript text).
- **aixiv.science** assisted manuscript quality review (clarity, structure, and reviewer-style feedback used to guide revisions).

AI did not invent experimental numbers. Fitting, residual scoring, baselines, and hybrid GPU search are conventional Rust / PyTorch with deterministic logs.

Literature and reproducibility artifacts. Open literature was also gathered with local OpenAlex / Crossref / arXiv providers. Every bibliography entry has a downloadable OA PDF (literature PDFs). Audible demos: hear-sample WAVs.

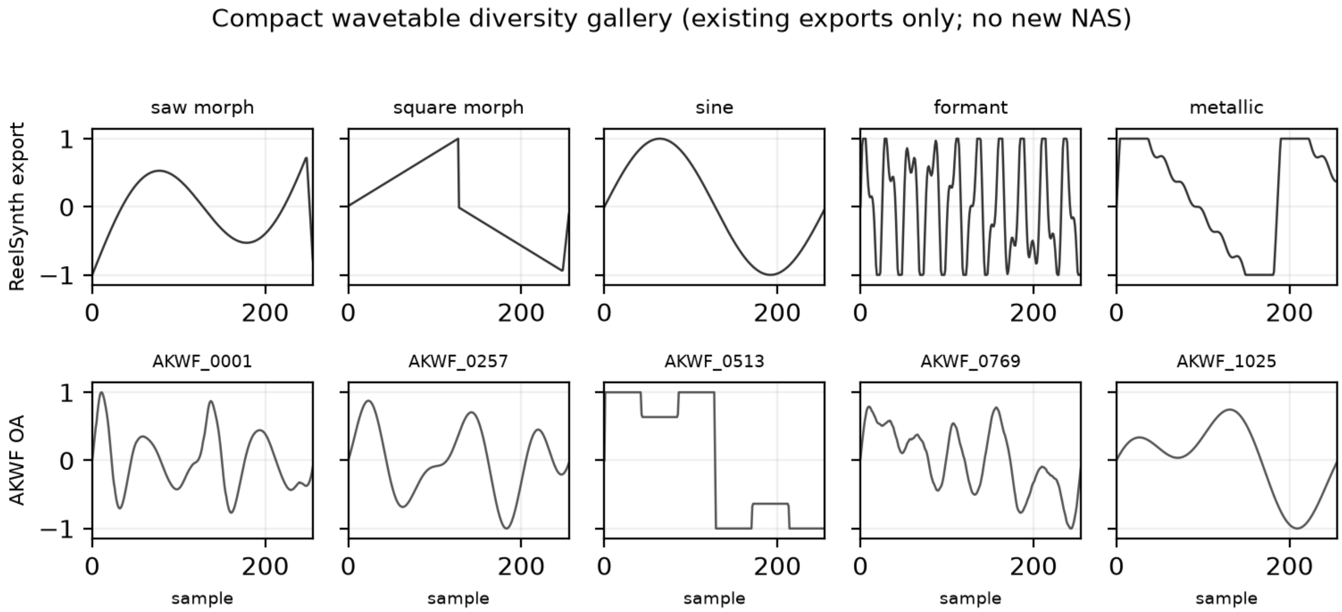

Figure 19: Compact wavetable diversity gallery from existing exports (one mid-morph dry frame per unique ReelSynth factory bank, top, and matched AKWF CC0 cycles, bottom). No new NAS for this panel. Quantitative wrap scores: Table 12.

Table 15: Wrap-heal transfer under locked R_{blend} (higher better; wrap-protocol stress test, not domain SOTA). DenoiseOpt: `hybrid_lstm` per domain (seed `1902771841`). Boards: CWRU/MFPT/Paderborn K001 bearing revolutions, MIT-BIH/PTB-XL ECG beats, synthetic CNC/PMU pilots (n=256 periods each). SeamN2N: corrupt→corrupt holdout (4000 steps, seed `424242`). Cycle-GAN / BeatDiff: author weights, OOD wrap probe on ECG only (— elsewhere).

Method	CWRU	MFPT	Paderborn	MIT-BIH	PTB-XL	synth CNC	synth PMU
Ours (`hybrid_lstm`)	0.9596	0.9427	0.9369	0.9747	0.9451	0.9914	0.9919
SeamN2N (ours, N2N-style)	0.8660	0.8662	0.8421	0.7495	0.7537	0.5534	0.9589
no-bake	0.8451	0.8657	0.8376	0.7495	0.7617	0.4499	0.9356
endpoint-pin	0.5054	0.5446	0.5670	0.8650	0.8696	0.4096	0.9442
`seam_fir3`	0.4483	0.5726	0.5601	0.7662	0.7604	0.5344	0.9119
DualCosine	0.4608	0.4830	0.4710	0.4137	0.5118	0.4943	0.8272
BeatDiff (author, OOD)‡	—	—	—	0.4907	0.4933	—	—
Cycle-GAN (author, OOD)	—	—	—	0.1063	0.2105	—	—
Al Firdausi backbone (wrap bake)†	—	—	0.8415	—	—	—	—

†Architecture-reuse residual bake on author CNN_1D_2L conv blocks (not the published fault classifier, Paderborn only). ‡Drive Orbax prior (step 9360), one-step denoise at σ=0.5 after mono→9-lead tile. Not Bedin native metrics.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument workflows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 19). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seeds: `20260719`–`20260723`. Search seeds: `1902771841`, `2026072701`, `2026072702`. Noise2Noise train seed: `424242`. Primary code lives at `https://github.com/reeldemo/reelsynth` (search runner: `overnight_gpu_rl_arch.py`, matched-5k bake-off: `bench_meta_approaches_5k.py`, SOTA bench: `bench_sota_matrix.py`, VA seam: `va_seam_blep.py`), with paper companion at `https://github.com/reeldemo/denoise-opt-meta`. CUDA benches need Python 3.12, PyTorch 2.6+CUDA 12.4, and NumPy/matplotlib; Rust/`cargo` builds the separate engine hardness probe. Frozen numbers live

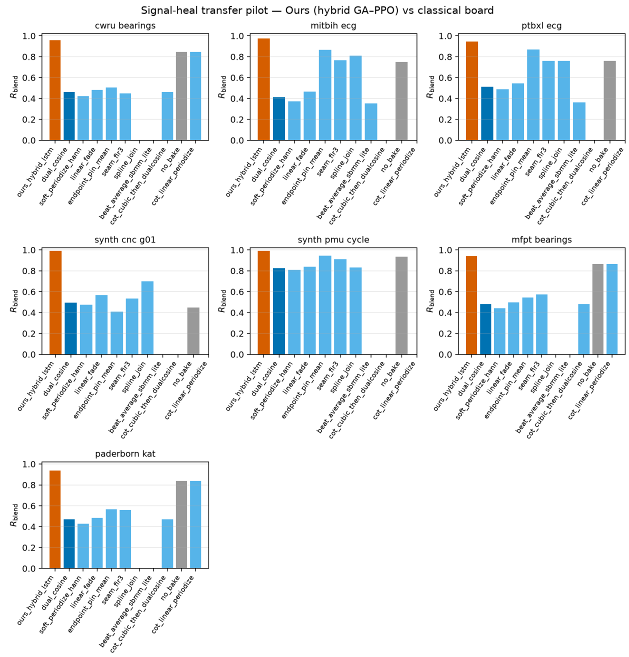

Figure 20: Grouped bars for Table 15: R_{blend} by board and method (higher better). Core rows (Ours, SeamN2N, no-bake, endpoint-pin, `seam_fir3`, DualCosine) appear on all seven boards. Hatched bars are author OOD wrap probes on ECG (Beat-Diff, Cycle-GAN) and the Paderborn Al Firdausi architecture-reuse wrap bake (not the published fault classifier, see Table 16). Missing cells are omitted rather than zero-filled.

under brand/artifacts (e.g. `sota_matrix.json`, `meta_approach_compare_v13_rblend/`, `multiseed_summary.json`). PESQ/STOI are omitted for non-speech cycles (domain mismatch), and MUSHRA was not run.

8.1 Funding

This research received no external funding.

8.2 Author contributions (CRediT)

Julian M. Kleber: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing (original draft), Writing (review and editing), Visualization.

8.3 Data availability

Frozen evaluation artifacts (JSON matrices, search histories, fitted champions) are published with the companion repositories: `https://github.com/reeldemo/reelsynth` (brand/artifacts) and `https://github.com/reeldemo/denoise-opt-meta` (this paper folder's `figures/` plus aggregate JSONs). Tables regenerate from those files via the named bench scripts; figures regenerate with `bench_meta_approaches_5k.py -aggregate-only` and `plot_paper_dataset_and_comparison_figures.py`; the PDF rebuilds with `build.ps1` in this folder. No private studio recordings were used.

8.4 Competing interests

None declared.

9 Limitations

We collect honesty notes here rather than repeating them through Results and Discussion.

- **Not a perceptual score.** R_{blend} and J use a procedural no-cliff sibling for seams and an engine-identity prior for the mid-cycle body. They do not measure listening quality. No formal MOS or MUSHRA was run. Downloadable WAVs and vibrato spectrograms support informal A/B only (Section 5.21). PESQ and STOI are excluded: primary cycles are non-speech wavetable tiles. We do not invent THD+N tables. Secondary SNR/SDR already appear on some strata boards (e.g. Tables 9, 13).

- **Metric lock.** Primary claims use the locked v10.1 R_{blend} protocol (Table 6 and Section 3.3). Whole-curve prolonged R near 0.991 (including the historical freeze $R \approx 0.99093$ vs Dual Cosine ≈ 0.81658) is a superseded debug score. It is not a current holdout result and must not overwrite the gate reading (Ours below Noise2Noise on mean holdout R_{blend}).

- **Multi-seed protocol.** Holdout tables report mean±std over five evaluation seeds (`20260719`–`20260723`). Matched-5k outer-loop re-search under R_{blend} is complete across three search seeds (`1902771841`, `2026072701`, `2026072702`; Table 17). Only one of three hybrid seeds clears the frozen Noise2Noise gate.

Table 16: Author-method and data-cut status under this wrap-protocol stress test (no invented scores).

Scope	Note
SeamN2N (ours, N2N-style)	Scored on all seven Table 15 boards under R_{blend}. Not Lehtinen code
Cycle-GAN (ECG, author)	OOD wrap probe in Table 15 (MIT-BIH 0.1063, PTB-XL 0.2105). Clinical restore $\neq$ wrap residual
BeatDiff (Bedin NeurIPS 2024)	OOD wrap probe in Table 15 (MIT-BIH 0.4907, PTB-XL 0.4933). Drive prior (not HF). Clinical morph prior $\neq$ wrap residual
Paderborn KAt deep (Al Firdausi CNN)	Trained CNN_1D_2L from scratch (not frozen `model.pth`) on bearings K001/KA04/KI04/KB23. File-holdout seed `20260726`, train seed `42`, 20 epochs. 4-class holdout acc 0.8590 (n=2560 segs, NORMAL/IR/OR/OR+IR). Classifier wrap residual N/A. Arch-reuse wrap bake listed in Table 15†. Wrap board: Ours 0.9369 / SeamN2N 0.8421 / DualCosine 0.4710
Expanded PTB-XL	records500 lead-I expanded (∼2k + `_hr` records, board n=256, Ours 0.9451). Multi-lead still out of scope
Real KIT CNC / IEEE PMU	Synthetic pilots only (download walls)
Formal MOS / MUSHRA	Not collected

Sign tests / win-rates on multi-family boards use paired per-family scores.

- **Domain Noise2Noise vs transfer depth.** Domain-trained Noise2Noise quality *was* run on the seven wrap-protocol stress-test boards (Table 15). It trails Ours on every column under R_{blend}. Latency timings in Appendix B still use the wavetable checkpoint zero-shot. Cycle-GAN / BeatDiff / real KIT–PMU / MOS remain incomplete where noted (Table 16).

- **Domain shift on exports.** On Factory+FX / Rust tiles the meta favorite can trail no-bake or FIR while still beating Dual Cosine. We do not claim universal factory superiority.

- **Search geometry.** Search scores Python generator draws. The Python multi-family matrix is primary. Rust export is secondary. Per-trial width is capped. Demucs-/diffusion-scale stacks were screened out.

- **No wrap-closure theorems.** Endpoint-pin can zero wrap-jump while residual falls. We claim tiled residual repair toward $r^{\star}$, not endpoint closure or search convergence.

- **Transfer scope.** Sci/eng wrap transfer is a classical-board stress test of the wrap protocol, not deep CWRU/ECG SOTA (Table 16).

- **Hyperparameter probe.** The ±50% one-at-a-time probe (Appendix B) is not a full re-search of Table 2.

9.1 Future work

A follow-up can treat which sounds fail as the primary object: stratified residual heatmaps over procedural families, cliff-size conditioning, and family-specialist operators under the same outer loop. Formal listening (even a small author-plus-colleague preference note on high-wrap tiles) remains the clearest missing perceptual check.

10 Conclusion

DenoiseOpt casts wavetable wrap discontinuity as cycle-local seam repair under blended residual R_{blend} and latency-aware objective J. A hybrid GA+PPO(+PBT) outer loop is applied to compact repair operators for this DSP task. Noise2Noise is the neural gate. Dual Cosine is the classical fade baseline. Across five holdout seeds the hybrid freeze reaches R_{blend}=0.971±0.001 but does not clear Noise2Noise (0.977±0.001), while Dual Cosine sits at 0.542±0.038. On multi-family boards Ours edges Noise2Noise on mean-of-means. On hard cliffs classical fades collapse while searched operators hold. Transfer boards reuse the same wrap protocol as stress tests, not domain diagnosis SOTA. The contribution is searchable repair operators for cycle-local DSP and synthesis seam restoration. Code: https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta.

Acknowledgments

Compute for the CUDA hybrid search campaign used a local NVIDIA GeForce RTX 3090 workstation. Open literature was gathered with local OpenAlex / Crossref / arXiv providers and author PDF checks under open-access constraints (see Section 7). No external collaborators contributed text, code, or experimental numbers for this preprint.

A Algorithms

Companion pseudocode for the residual score, classical controls, FitCell, and hybrid outer-loop branches. Canonical narrative lives in Section 3. These listings mirror the pseudocode notes and the CUDA search runner.

B Supplementary tables and figures

Material moved here for the conference-length reading path: matched outer-loop bake-off, branch/compute freezes, hyperparameter probe, transfer latency, and search-monitor plots. Primary claims remain in Sections 5–5.20.

Algorithm 1 ResidualScore

Require: Ideal cycle $r^\star$, baked cycle y, periods N

$I \leftarrow \mathrm{Tile}(r^\star, N)$
$Y \leftarrow \mathrm{Tile}(y, N)$
$r_{\mathrm{rms}} \leftarrow \mathrm{RMS}(Y - I)$
$i_{\mathrm{rms}} \leftarrow \max(\mathrm{RMS}(I), \varepsilon)$
return $\mathrm{clamp}(1 - r_{\mathrm{rms}}/i_{\mathrm{rms}}, 0, 1)$

Algorithm 2 DualCosine baseline

Require: Engine cycle x

Fade both ends of x with raised-cosine windows of width W
return ResidualScore($r^\star$, x_{faded}, N)

Algorithm 3 No-bake (passthrough) control

Require: Engine cycle x (open-wrap cliff present)

return ResidualScore($r^\star$, x, N) ▷ no operator: $x \mapsto x$

B.1 Matched-budget outer-loop comparison

Under a matched 5,000-evaluation budget with LSTM and xLSTM in the searchable bake vocabulary, how do Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS compare to Ours (hybrid GA+PPO)? Table 17 reports the completed Deferred D1 matched re-search under locked $R_{\mathrm{blend}}+J$ (mean±std over three search seeds; artifacts `meta_approach_compare_v13_rblend/`). Ours leads (R_{blend}=0.9748±0.0018; mean wall ≈8.4 h). Aging evolution is second (0.9635±0.0049); Arch REINFORCE is highest-variance (0.9599±0.0116). Only seed `1902771841` hybrid clears the frozen Noise2Noise gate (R_{blend}≈0.9766 vs gate ≈0.9750). Hybrid live champions stay compact (≈24k±2k params across seeds) despite mid-search graphs that swing from ≈15k to ≈370k. Noise2Noise is not an outer-loop method here (see Tables 6 and 9). Learning-curve and bar figures use seed `1902771841` trajectories under the same locked objective.

B.2 Ablations and compute

Table 18 reports branch-best freezes at the 5k gate from one hybrid history (prolonged R; no separate fitted weights), plus isolated 150-iteration `ablate-*` final champions re-scored under R_{blend}. Table 19 summarizes campaign compute (historical prolonged-R overnight).

B.3 Hyperparameter sensitivity

One-at-a-time ±50% probe of Table 2 knobs at 500 outer iterations (seed `1902771841`). Not a full 5k re-search. Default champion R≈0.9888. The largest $|\Delta R|$ among completed arms is ≈0.0123 (50% lower learning rate). All 11/11 arms completed.

B.4 Transfer inference latency

Table 21 and Figure 24 pair with the prolonged-R transfer board (Table 15). N2N latency uses the wavetable checkpoint zero-shot. Domain-trained N2N quality scores are in Table 15.

Algorithm 4 MoE soft gating over bake experts

Require: Cycle x, expert operators $\{E_k\}_{k=1}^{K}$, gate logits $g \in \mathbb{R}^K$

$w \leftarrow \mathrm{softmax}(g)$
$y \leftarrow \sum_{k=1}^{K} w_k E_k(x)$
return y

Algorithm 5 FitCell (inner loop)

Require: Architecture cfg, fit steps T, batch size B

cell ← SeamCell(cfg)
opt ← Adam(cell.params)
prev ← null
patience ← 0
for $t = 1, \ldots, T$ **do**
  ideal, eng ← MakeBatch(B)
  out ← ApplyOps(eng, cell, cfg.ops)
  $R \leftarrow$ mean ResidualScore(ideal, out)
  loss ← $1 - R$
  if finite(loss) **then**
    AdamStep(opt, loss)
  end if
  if prev ≠ null and $|\mathrm{prev} - R| / \max(|\mathrm{prev}|, \varepsilon) < 10^{-4}$ **then**
    patience ← patience +1
    if patience ≥ 3 **then**
      break
    end if
  else
    patience ← 0
  end if
  prev ← R
end for
return cell, R

Algorithm 6 GA tournament step

Require: Population P, tournament size k, crossover rate p_c, mutation rate p_m

Sample k genomes; parents ← top-2 by R
child ← Crossover(parents) with prob. p_c, else clone elite
Mutate(child) with rate p_m (ops, depth, operator type, θ)
Fit and evaluate child; insert into P (replace worst if full)
return updated P

B.5 Search-monitor figures

C Transfer artifact locations

Full cross-domain wrap-heal transfer protocol, classical-board tables, and deep-model status rows are in Sections 4.2–5.20 of the main text. Algorithms for the residual score and hybrid loop remain in Appendix A. Raw JSON and figures: signal_heal_transfer.

Algorithm 7 PPO architecture update

Require: Actor–critic π_ϕ, clip ε, experience buffer of mutate actions

Sample action a editing ops / hyperparameters from π_ϕ
Fit proposed cfg; observe centered advantage $\rho = R - R_{\text{DualCosine}}$ ▷ objective remains maximize R
Advantage $\hat{A}$ from critic; ratio $r_\phi = \pi_\phi(a)/\pi_{\phi_{\text{old}}}(a)$
Update ϕ with clipped surrogate $\min(r_\phi \hat{A}, \text{clip}(r_\phi, 1-\varepsilon, 1+\varepsilon)\hat{A})$
return updated π_ϕ

Algorithm 8 PBT exploit–mutate

Require: Population P, exploit quantile q, mutate scale σ

Rank P by R; for each bottom member, copy hyperparameters from a top-q elite
Perturb copied hyperparameters by multiplicative noise of scale σ
Refit briefly; write back into P
return updated P

Algorithm 9 HybridMetaSearch

Require: Iteration budget T_{out}, population size n

```
P ← InitPopulation(n)
π ← ActorCritic()
champ ← null
boredom ← 0
for it = 1, ..., T_out do
    branch ← Rotate(ppo, ga, pbt, nas, combo)
    cfg, hp ← Propose(branch, P, π)
    cell, R_fit ← FitCell(cfg, hp.fit_steps)
    R_eval ← EvalCell(cell)
    R ← ½R_fit + ½R_eval (+ optional depth/MoE bonus)
    UpdatePopulation(P, cfg, R)
    PPOUpdate(π, centered ρ = R − R_DualCosine)
    if R > champ.R then
        champ ← (cfg, cell, R)
        boredom ← 0
    else
        boredom ← boredom + 1
    end if
    if boredom ≥ 1000 then
        PlateauAdapt
        boredom ← 0
    end if
end for
return champ
```

Table 17: Matched 5,000-evaluation outer-loop comparison under locked $R_{\text{blend}}+J$ (Deferred D1 complete). Mean±std over three search seeds (`1902771841`, `2026072701`, `2026072702`). ΔR uses seed-`1902771841` Dual Cosine (≈0.504). Wall-h is mean CUDA wall per approach. LSTM?/xLSTM?: whether any seed's champion included that block.

Method	R_{blend}	ΔR vs DC	h	LSTM	xLSTM
Random NAS	0.9367±0.0032	+0.4327	4.7	no	no
Cont. CMA-ES	0.9565±0.0016	+0.4525	1.8	no	no
Arch REINFORCE	0.9599±0.0116	+0.4559	5.8	yes	no
Aging evolution	0.9635±0.0049	+0.4595	9.0	yes	no
TPE Bayes NAS	0.9451±0.0019	+0.4411	3.8	no	no
Ours (hybrid GA–PPO)	0.9748±0.0018	+0.4708	8.4	no	no

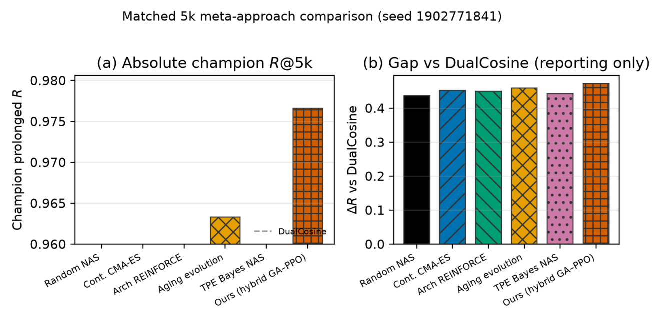

Figure 21: Matched 5,000-evaluation search methods under locked $R_{\text{blend}}+J$ (seed `1902771841`). Table 17 reports mean±std across three search seeds. Dashed line: Dual Cosine on the same runner.

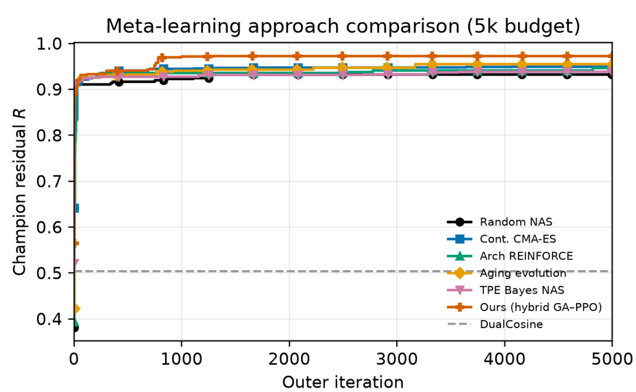

Figure 22: Champion R_{blend} over a matched 5,000-evaluation budget (Deferred D1 complete; seed `1902771841`). Dashed line: Dual Cosine.

Table 18: Ablations. Left: best prolonged R by branch in the recorded search run (final iteration ≈6,922; superseded search metric). Right: isolated 150-iteration `ablate-*` final champions re-scored under R_{blend} (α=0.7; seed `1902771841`). Full R_{blend} branch re-search remains deferred.

Config	Search freeze prolonged R	Isolated 150-it R_{blend}
Full hybrid	0.99093	0.92710
GA-only	0.99016	0.94168
PBT (search branch)	0.99012	n/a
PPO-only	0.98924	0.94405
GA+PPO	n/a	0.94478
Combined branch	0.98854	n/a
NAS branch	0.98677	n/a
Dual Cosine (runner)	0.81658	≈0.539

Table 19: Compute budget for the reported 5,000-evaluation search run (RTX 3090, search seed `1902771841`; prolonged-R objective). Final champion R below is the historical prolonged freeze, not R_{blend}.

Item	[illegible]
GPU	NVIDIA GeForce RTX [illegible]
Elapsed wall time	≈[illegible]
Clean iterations / arch evals	≈[illegible]
Population size	[illegible]
Final champion prolonged R	0.99093
Dual Cosine baseline (runner, prolonged)	0.81658
Peak VRAM (replay)	≈3.29 GiB

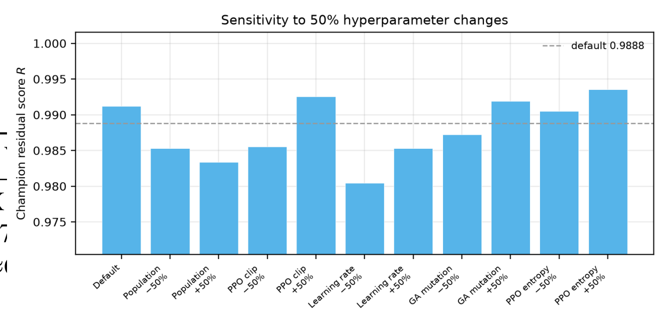

Figure 23: One-at-a-time ±50% HP sensitivity of hybrid champion R at 500 iterations. Blue bar: Table 2 default.

Table 20: One-at-a-time ±50% hyperparameter sensitivity probe (500 outer iterations, seed `1902771841`, sine+cliff FitCell). Champion absolute R is compared with the Table 2 default. This is sensitivity evidence only, not a full 5,000-iteration re-search for each parameter.

Parameter change	Champ. R	ΔR	Iter.
Default	0.98882	+0.00000	500
Population size −50%	0.98136	-0.00746	500
Population size +50%	0.98073	-0.00809	500
PPO clip −50%	0.98159	-0.00723	500
PPO clip +50%	0.99041	+0.00159	500
Learning rate −50%	0.97654	-0.01228	500
Learning rate +50%	0.98138	-0.00744	500
GA mutation rate −50%	0.98334	-0.00548	500
GA mutation rate +50%	0.99002	+0.00120	500
PPO entropy −50%	0.98912	+0.00030	500
PPO entropy +50%	0.99116	+0.00234	500

Table 21: Transfer-domain inference latency (ms/batch). Batch=64, CUDA.

Method	CWRU	MFPT	MIT-BIH	PTB-XL	CNC	PMU
no-bake	0.26	0.26	0.27	0.27	0.28	0.26
Dual Cosine	1.10	1.11	1.12	1.14	1.15	1.14
Ours (hybrid)	3.92	6.61	2.87	1.67	1.69	1.67
N2N (WT to domain)	0.95	0.99	0.98	0.97	0.98	0.97

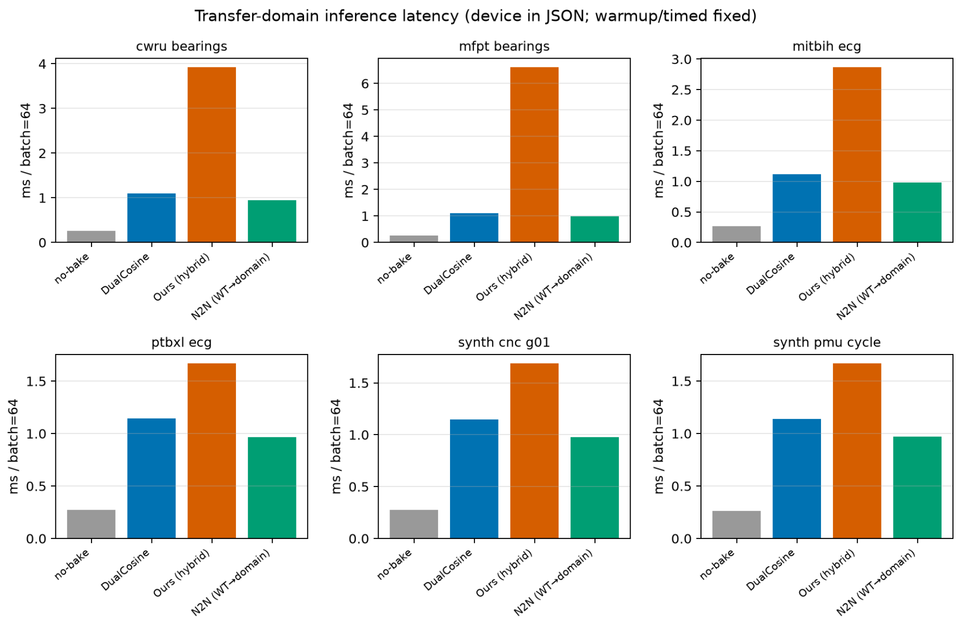

Figure 24: Transfer-domain inference latency (ms/batch). N2N is wavetable-trained zero-shot when available.

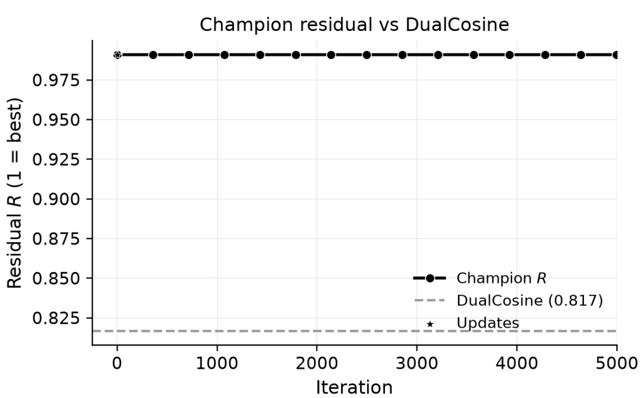

Figure 25: Champion residual R over the first 5,000 search iterations (Dual Cosine dashed).

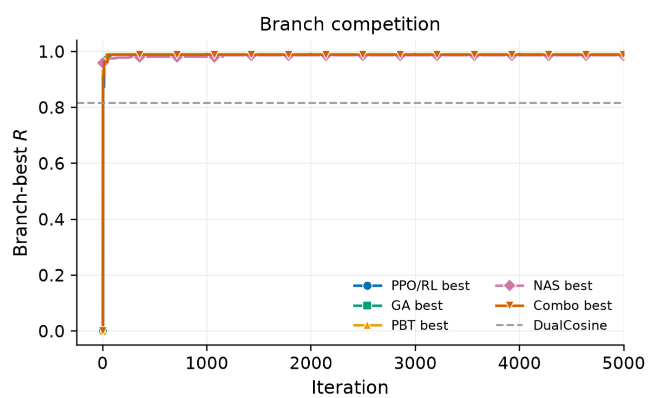

Figure 26: Best residual by search branch over the first 5,000 iterations.

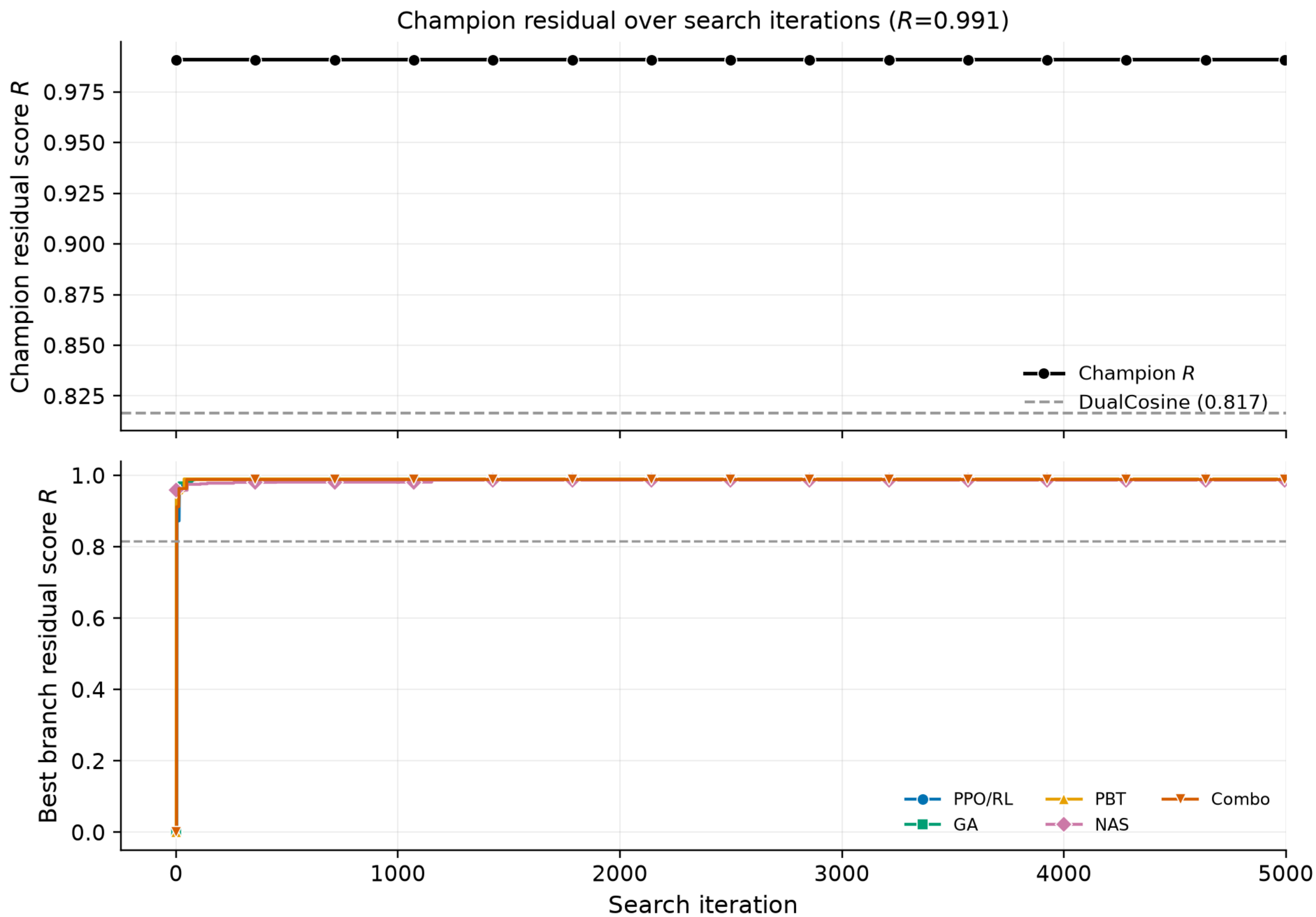

Figure 27: Search progress: champion R (top) and branch bests (bottom).

Figure 28: Per-trial residual by branch with champion trajectory overlaid.

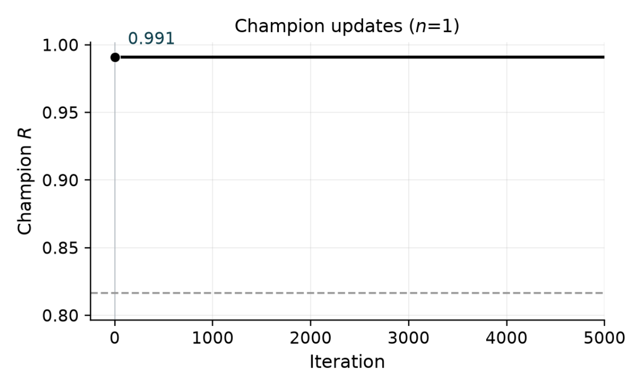

Figure 29: Champion updates over the first 5,000 iterations.

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